

Лабораторная работа №8

Адресация IPv4 и IPv6. Настройка маршрутизации

Комягин Андрей Николаевич

Содержание

1	Цель работы	6
2	Задание	7
3	Выполнение лабораторной работы	8
3.1	Документирование сети	8
3.2	Построение стенда в GNS3	13
3.3	Настройка IPv4/IPv6 адресации	18
3.3.1	Настройка адресов на PC1 и PC2 (VPCS)	18
3.3.2	Настройка адресов на маршрутизаторах (FRR)	23
3.4	Динамическая маршрутизация RIP/RIPng	30
3.4.1	RIP для IPv4	30
3.4.2	RIPng для IPv6	40
3.5	Динамическая маршрутизация OSPF (OSPFv2/OSPFv3)	49
3.5.1	OSPFv2 для IPv4	49
4	Выводы	55

Список иллюстраций

3.1	Топология в GNS3 (общий вид стенда)	10
3.2	Схема L1 (физические соединения и интерфейсы)	11
3.3	Схема L3 (адресация и направление маршрутизации)	12
3.4	Таблицы/схемы адресации, используемые в отчёте	13
3.5	Создание проекта и размещение устройств в GNS3	14
3.6	Переименование узлов и настройка отображаемых имён	15
3.7	Включение захвата трафика на заданных линках	16
3.8	Проверка интерфейсов/соединений в собранной топологии	17
3.9	Окна захвата трафика (подготовка к анализу протоколов)	17
3.10	Настройка IPv4 на PC1 и проверка show ip	18
3.11	Настройка IPv4 на PC2 и проверка show ip	19
3.12	Настройка IPv6 на PC1 и проверка show ipv6	21
3.13	Настройка IPv6 на PC2 и проверка show ipv6	22
3.14	Настройка IPv4 на msk-ankomyagin-gw-01 (eth0/eth1/eth2)	23
3.15	Настройка IPv4 на msk-ankomyagin-gw-02 (eth0/eth1)	24
3.16	Настройка IPv4 на msk-ankomyagin-gw-03 (eth0/eth1/eth2)	25
3.17	Настройка IPv4 на msk-ankomyagin-gw-04 (eth0/eth1)	26
3.18	Настройка IPv6 на msk-ankomyagin-gw-01 и параметры ND/RA на eth0	27
3.19	Настройка IPv6 на msk-ankomyagin-gw-02	27
3.20	Настройка IPv6 на msk-ankomyagin-gw-03 и параметры ND/RA на eth0	28
3.21	Настройка IPv6 на msk-ankomyagin-gw-04	29
3.22	Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-01	30
3.23	Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-02	31
3.24	Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-03	32
3.25	Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-04	33
3.26	Проверка таблицы RIP-маршрутов (show ip route rip)	34
3.27	Ping PC1 → PC2 (IPv4) после включения RIP	35
3.28	Trace PC1 → PC2 (IPv4) и определение пути	36
3.29	Отключение eth0 на msk-ankomyagin-gw-02 для проверки перестроения маршрута	37
3.30	Повторный ping/trace после отключения линка (IPv4, RIP)	38
3.31	Захват трафика RIP на линке sw-01—gw-01	39
3.32	Захват трафика RIP на линке sw-02—gw-03	40
3.33	Конфигурация RIPng на msk-ankomyagin-gw-01	41
3.34	Конфигурация RIPng на msk-ankomyagin-gw-02	42
3.35	Конфигурация RIPng на msk-ankomyagin-gw-03	43
3.36	Конфигурация RIPng на msk-ankomyagin-gw-04	44

3.37 Ping PC1 → PC2 (IPv6) после включения RIPng	45
3.38 Trace PC1 → PC2 (IPv6) и определение пути	46
3.39 Отключение eth0 на msk-ankomyagin-gw-02 (IPv6, RIPng)	47
3.40 Повторный ping/trace после отказа линка (IPv6, RIPng)	47
3.41 Захват трафика RIPng на линке sw-01—gw-01	48
3.42 Захват трафика RIPng на линке sw-02—gw-03	48
3.43 Конфигурация OSPFv2 на msk-ankomyagin-gw-01 (area 0)	49
3.44 Конфигурация OSPFv2 на msk-ankomyagin-gw-02 (area 0)	50
3.45 Конфигурация OSPFv2 на msk-ankomyagin-gw-03 (area 0)	51
3.46 Конфигурация OSPFv2 на msk-ankomyagin-gw-04 (area 0)	51
3.47 Проверка соседей OSPFv2 (show ip ospf neighbor)	52
3.48 Проверка маршрутов OSPFv2 (show ip ospf route)	53
3.49 Ping/trace (IPv4) при OSPFv2 до/после отказа линка	53
3.50 Захват трафика OSPF/служебных пакетов на контролируемых линках	54

Список таблиц

1 Цель работы

Цель работы — изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

2 Задание

Требуется задокументировать моделируемую сеть (схемы L1 и L3, таблицы адресации), настроить двойной стек IPv4/IPv6, а также настроить динамическую маршрутизацию по протоколам RIP/RIPng и OSPF (OSPFv2/OSPFv3), проверить связность, пути прохождения пакетов и проанализировать захваченный трафик.

3 Выполнение лабораторной работы

3.1 Документирование сети

Ниже приведены таблицы адресации (сети и адреса интерфейсов), используемые в работе.

Таблица: Адреса сегментов сети (по табл. 8.1).

Сегмент	Сеть IPv4	Сеть IPv6
PC1 – gw-01	10.0.10.0/24	2001:10::/64
PC2 – gw-03	10.0.11.0/24	2001:11::/64
gw-01 – gw-02	10.0.1.0/24	2001:1::/64
gw-02 – gw-03	10.0.2.0/24	2001:2::/64
gw-03 – gw-04	10.0.3.0/24	2001:3::/64
gw-04 – gw-01	10.0.4.0/24	2001:4::/64

Таблица: Адреса интерфейсов устройств (по табл. 8.2).

Устройство	Интерфейс	Адрес IP/префикс	Шлюз по умолчанию	Следующее устройство
gw-01	eth0	10.0.10.1/24	n/a	PC1
gw-01	eth0	2001:10::1/64	n/a	PC1
gw-01	eth1	10.0.1.1/24	n/a	gw-02
gw-01	eth1	2001:1::1/64	n/a	gw-02

Устройство	Интерфейс	Адрес IP/префикс	Шлюз по умолчанию	Следующее устройство
gw-01	eth2	10.0.4.2/24	n/a	gw-04
gw-01	eth2	2001:4::2/64	n/a	gw-04
gw-02	eth0	10.0.1.2/24	n/a	gw-01
gw-02	eth0	2001:1::2/64	n/a	gw-01
gw-02	eth1	10.0.2.1/24	n/a	gw-03
gw-02	eth1	2001:2::1/64	n/a	gw-03
gw-03	eth0	10.0.11.1/24	n/a	PC2
gw-03	eth0	2001:11::1/64	n/a	PC2
gw-03	eth1	10.0.2.2/24	n/a	gw-02
gw-03	eth1	2001:2::2/64	n/a	gw-02
gw-03	eth2	10.0.3.1/24	n/a	gw-04
gw-03	eth2	2001:3::1/64	n/a	gw-04
gw-04	eth0	10.0.3.2/24	n/a	gw-03
gw-04	eth0	2001:3::2/64	n/a	gw-03
gw-04	eth1	10.0.4.1/24	n/a	gw-01
gw-04	eth1	2001:4::1/64	n/a	gw-01
PC1	NIC	10.0.10.10/24	10.0.10.1	gw-01
PC1	NIC	2001:10::a/64	n/a	gw-01
PC2	NIC	10.0.11.10/24	10.0.11.1	gw-03
PC2	NIC	2001:11::a/64	n/a	gw-03

В отчёте также подготовлены схемы L1 и L3 для данной топологии.

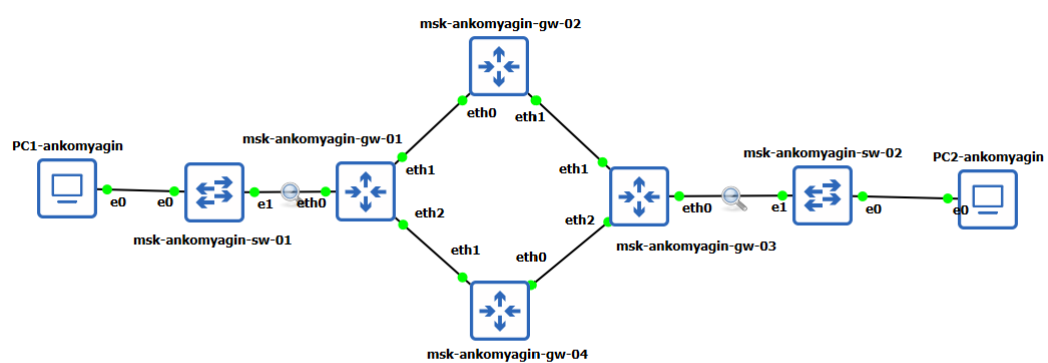
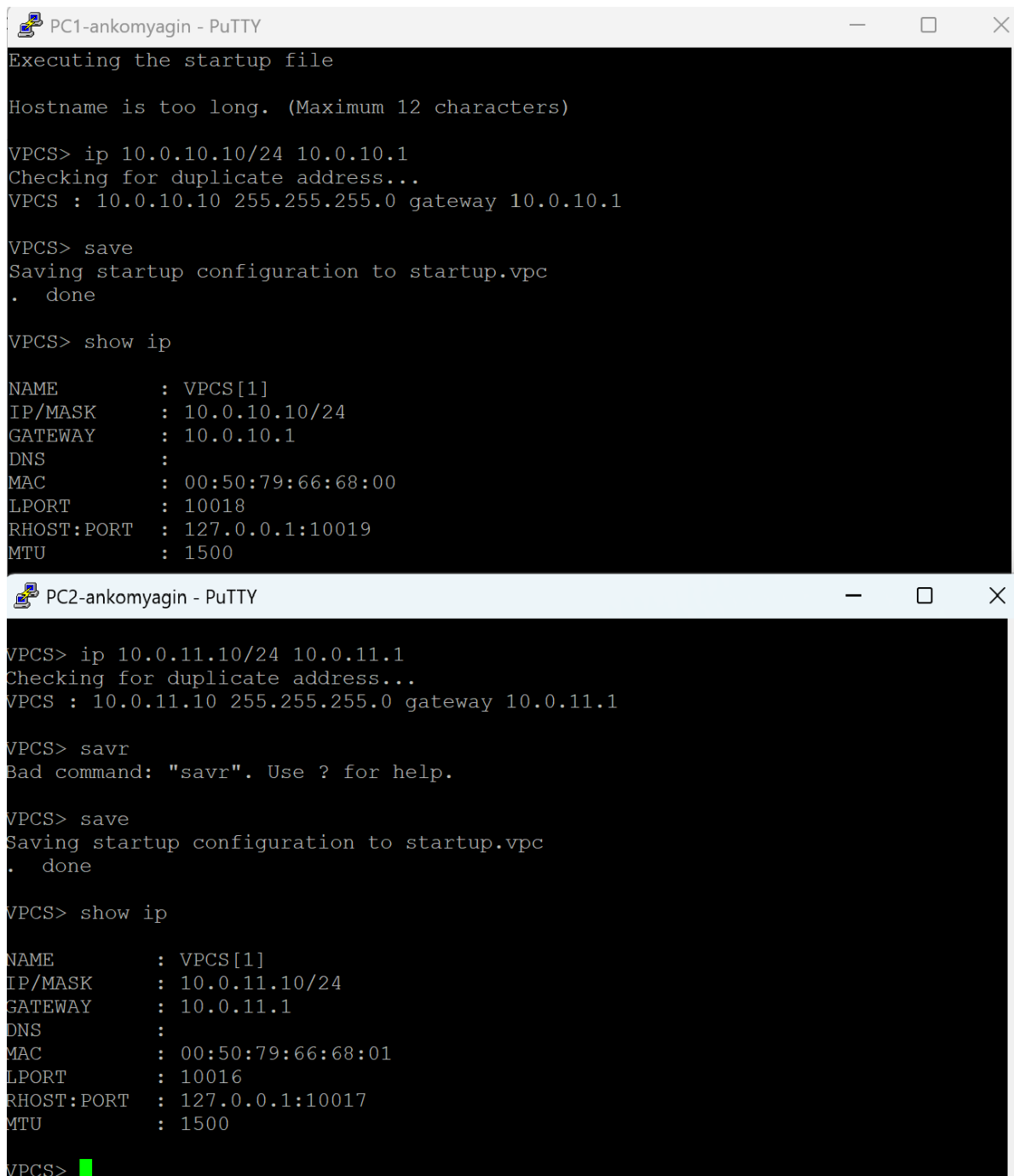


Рис. 3.1: Топология в GNS3 (общий вид стенда)



The image displays two screenshots of PuTTY terminal windows, labeled PC1 and PC2, showing the configuration of VPCS (Virtual Packet Capture System) for two different hosts.

PC1-ankomyagin - PuTTY

```
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 10.0.10.10/24
GATEWAY   : 10.0.10.1
DNS       :
MAC       : 00:50:79:66:68:00
LPORT     : 10018
RHOST:PORT : 127.0.0.1:10019
MTU       : 1500
```

PC2-ankomyagin - PuTTY

```
VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> savr
Bad command: "savr". Use ? for help.

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 10.0.11.10/24
GATEWAY   : 10.0.11.1
DNS       :
MAC       : 00:50:79:66:68:01
LPORT     : 10016
RHOST:PORT : 127.0.0.1:10017
MTU       : 1500

VPCS>
```

Рис. 3.2: Схема L1 (физические соединения и интерфейсы)

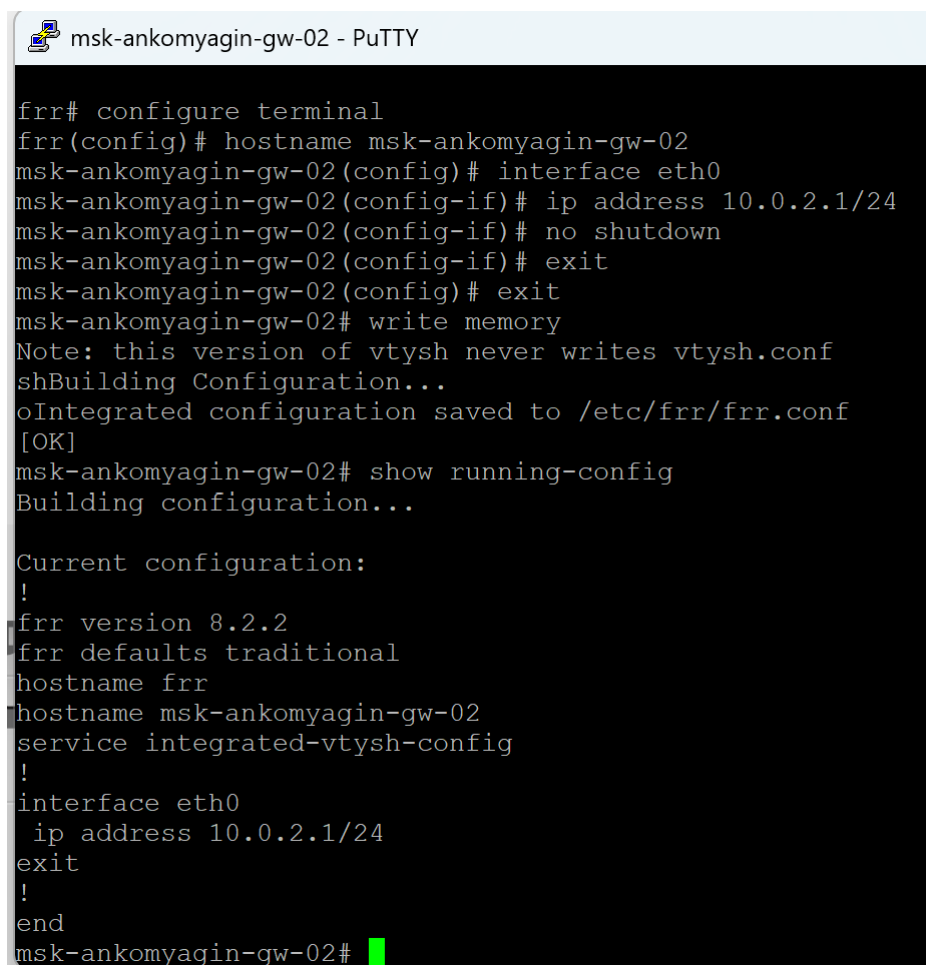
```
msk-ankomyagin-gw-01 - PuTTY
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-01
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ip address 10.0.10.1/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ip address 10.0.1.1/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth2
msk-ankomyagin-gw-01(config-if)# ip address 10.0.4.2/24
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-01
service integrated-vtysh-config
!
interface eth1
 ip address 10.0.1.1/24
 ip address 10.0.10.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-ankomyagin-gw-01#
```

Рис. 3.3: Схема L3 (адресация и направление маршрутизации)



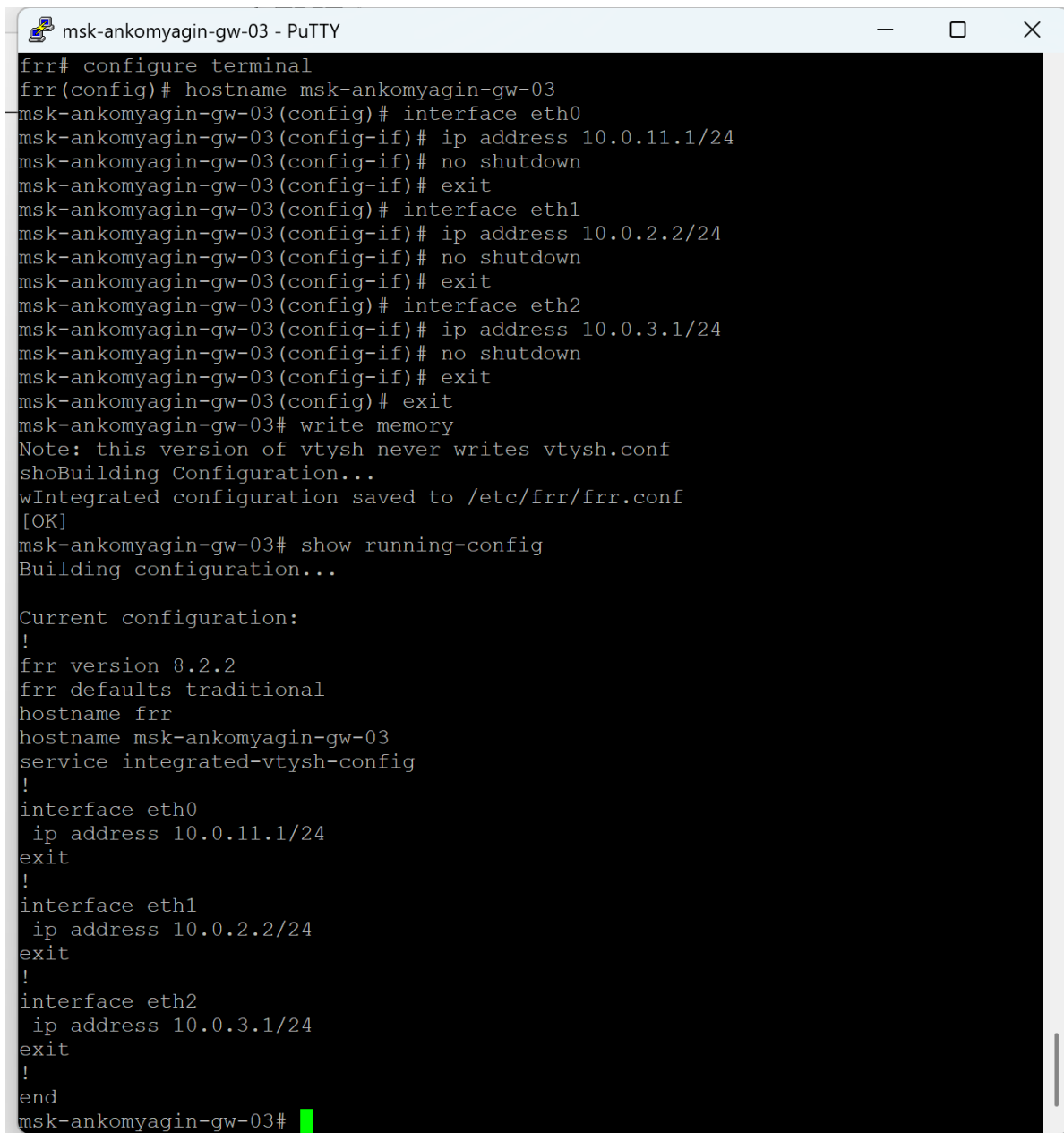
```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-02
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# ip address 10.0.2.1/24
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# exit
msk-ankomyagin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
shBuilding Configuration...
oIntegrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.2.1/24
exit
!
end
msk-ankomyagin-gw-02#
```

Рис. 3.4: Таблицы/схемы адресации, используемые в отчёте

3.2 Построение стенда в GNS3

В GNS3 был создан новый проект и собрана топология, соответствующая заданию: четыре маршрутизатора (FRR) и два хоста (VPCS), соединённые через коммутаторы согласно топологии сети.

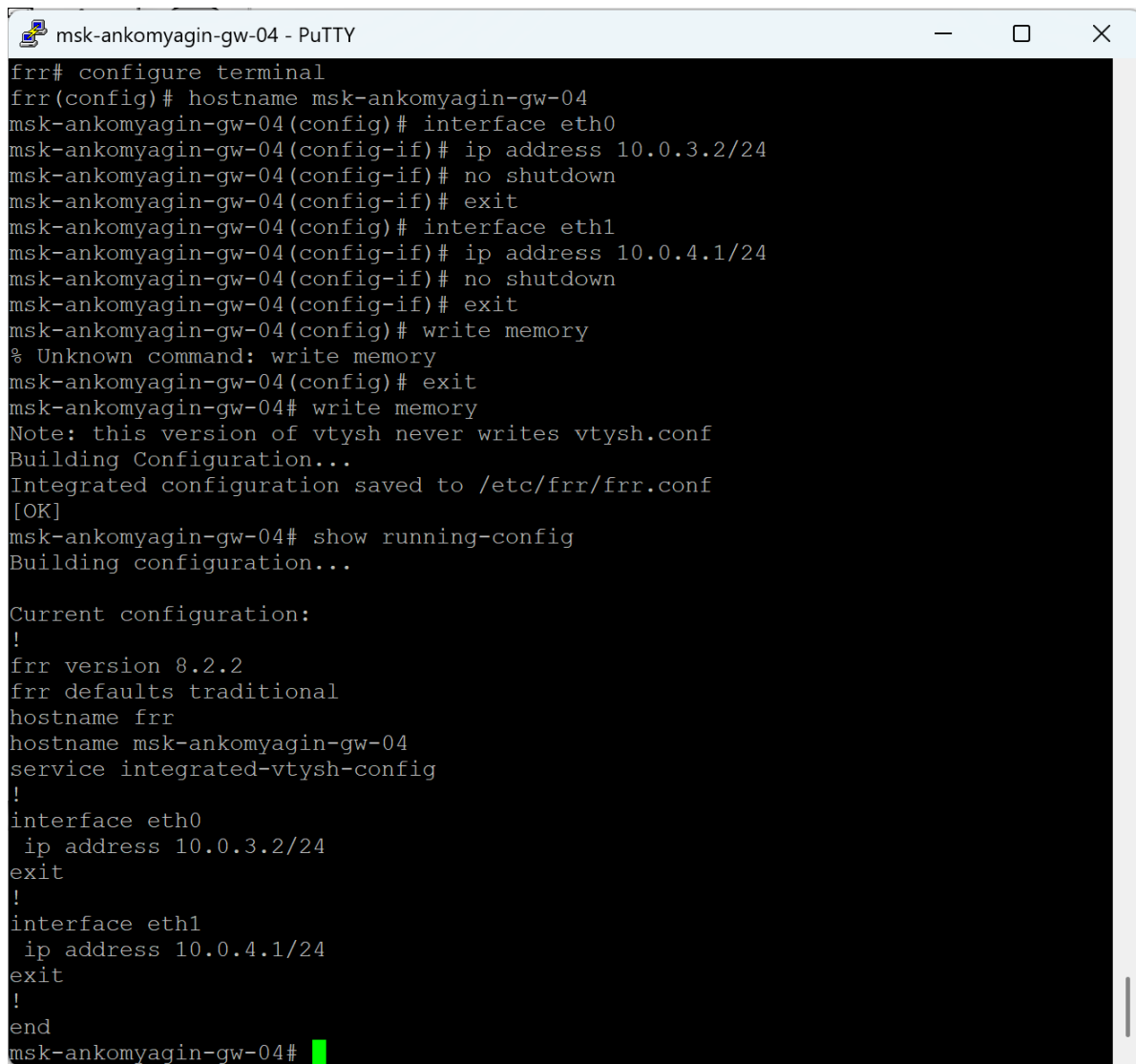


```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-03
msk-ankomyagin-gw-03(config)# interface eth0
msk-ankomyagin-gw-03(config-if)# ip address 10.0.11.1/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ip address 10.0.2.2/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth2
msk-ankomyagin-gw-03(config-if)# ip address 10.0.3.1/24
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
shoBuilding Configuration...
wIntegrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-ankomyagin-gw-03#
```

Рис. 3.5: Создание проекта и размещение устройств в GNS3

Имена устройств приведены к требуемому формату (msk-ankomyagin-sw-0x, msk-ankomyagin-gw-0x, PCx-ankomyagin), после чего включён захват трафика на соединениях sw-01—gw-01 и sw-02—gw-03.



```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-04
msk-ankomyagin-gw-04(config)# interface eth0
msk-ankomyagin-gw-04(config-if)# ip address 10.0.3.2/24
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ip address 10.0.4.1/24
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# write memory
% Unknown command: write memory
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-ankomyagin-gw-04#
```

Рис. 3.6: Переименование узлов и настройка отображаемых имён

The image displays two screenshots of a PuTTY terminal window titled "PC1-ankomyagin - PuTTY".

The top screenshot shows the configuration for PC1. It starts with a check for duplicate addresses, followed by setting the VPCS IP to 10.0.10.10 and the gateway to 10.0.10.1. Then, the IPv6 address 2001:10::a/64 is assigned to PC1. The configuration is saved to startup.vpc, and the "show ipv6" command is executed, displaying details for VPCS[1], including its link-local scope (fe80::250:79ff:fe66:6800/64), global scope (2001:10::a/64), MAC address (00:50:79:66:68:00), LPORT (10016), RHOST:PORT (127.0.0.1:10017), and MTU (1500).

The bottom screenshot shows the configuration for PC2. It follows a similar process: checking for duplicate addresses, setting the VPCS IP to 10.0.11.10 and the gateway to 10.0.11.1, assigning the IPv6 address 2001:11::a/64 to PC1, saving the configuration to startup.vpc, and executing the "show ip" command. The output shows details for VPCS[1], including its IP/MASK (10.0.11.10/24), gateway (10.0.11.1), MAC address (00:50:79:66:68:01), LPORT (10018), RHOST:PORT (127.0.0.1:10019), and MTU (1500).

Рис. 3.7: Включение захвата трафика на заданных линках

The image shows two terminal windows side-by-side. The left window is titled 'msk-ankomyagin-gw-03 - PuTTY' and shows the configuration of msk-ankomyagin-gw-03. The right window is titled 'msk-ankomyagin-gw-01 - PuTTY' and shows the configuration of msk-ankomyagin-gw-01. Both windows show the configuration of interfaces eth0, eth1, and eth2 with IPv4 and IPv6 addresses and prefixes. The configuration is saved to /etc/frr/frr.conf and the running configuration is shown.

```
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-ankomyagin-gw-03(config-if)# no ipv6 nd suppress-ra
msk-ankomyagin-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
% Unknown command: interface eth1
msk-ankomyagin-gw-03(config)# ipv6 address 2001:2::2/64
% Unknown command: ipv6 address 2001:2::2/64
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth2
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
show Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-03
service integrated-vtysh-config
!
interface eth0
ip address 10.0.11.1/24
ipv6 address 2001:11::1/64
ipv6 nd prefix 2001:11::/64
no ipv6 nd suppress-ra
exit
!
interface eth1
ip address 10.0.2.2/24
ipv6 address 2001:2::2/64
exit
!
interface eth2
ip address 10.0.3.1/24
ipv6 address 2001:3::1/64
exit
!
```

```
msk-ankomyagin-gw-01# configure terminal
msk-ankomyagin-gw-01(config)# ipv6 forwarding
msk-ankomyagin-gw-01(config)# interface eth0
msk-ankomyagin-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-ankomyagin-gw-01(config-if)# no ipv6 nd suppress-ra
msk-ankomyagin-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth2
msk-ankomyagin-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-ankomyagin-gw-01(config-if)# no shutdown
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
show Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-01
service integrated-vtysh-config
!
interface eth0
ip address 2001:10::1/64
ipv6 nd prefix 2001:10::/64
no ipv6 nd suppress-ra
exit
!
interface eth1
ip address 10.0.1.1/24
ip address 10.0.10.1/24
ipv6 address 2001:1::1/64
exit
!
interface eth2
ip address 10.0.4.2/24
ipv6 address 2001:4::2/64
exit
!
```

Рис. 3.8: Проверка интерфейсов/соединений в собранной топологии

The image shows two terminal windows side-by-side. The left window is titled 'msk-ankomyagin-gw-04 - PuTTY' and shows the configuration of msk-ankomyagin-gw-04. The right window is titled 'msk-ankomyagin-gw-02 - PuTTY' and shows the configuration of msk-ankomyagin-gw-02. Both windows show the configuration of interfaces eth0, eth1, and eth2 with IPv4 and IPv6 addresses and prefixes. The configuration is saved to /etc/frr/frr.conf and the running configuration is shown.

```
msk-ankomyagin-gw-04# configure terminal
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# write memory
% Unknown command: write memory
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-04
service integrated-vtysh-config
!
interface eth0
ip address 10.0.3.2/24
ipv6 address 2001:3::2/64
exit
!
interface eth1
ip address 10.0.4.1/24
ipv6 address 2001:4::1/64
exit
!
```

```
msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# ipv6 forwarding
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# interface eth1
msk-ankomyagin-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# exit
msk-ankomyagin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-ankomyagin-gw-02
service integrated-vtysh-config
!
interface eth0
ip address 10.0.2.1/24
ipv6 address 2001:1::2/64
exit
!
interface eth1
ip address 2001:2::1/64
exit
!
```

Рис. 3.9: Окна захвата трафика (подготовка к анализу протоколов)

3.3 Настройка IPv4/IPv6 адресации

3.3.1 Настройка адресов на PC1 и PC2 (VPCS)

Оконечным устройствам назначены IPv4-адреса командами VPCS в соответствии с табл. 8.2 (PC1: 10.0.10.10/24, шлюз 10.0.10.1; PC2: 10.0.11.10/24, шлюз 10.0.11.1).

```
PC1> ip 10.0.10.10/24 10.0.10.1
```

```
PC1> save
```

```
PC1> show ip
```

```
PC2> ip 10.0.11.10/24 10.0.11.1
```

```
PC2> save
```

```
PC2> show ip
```

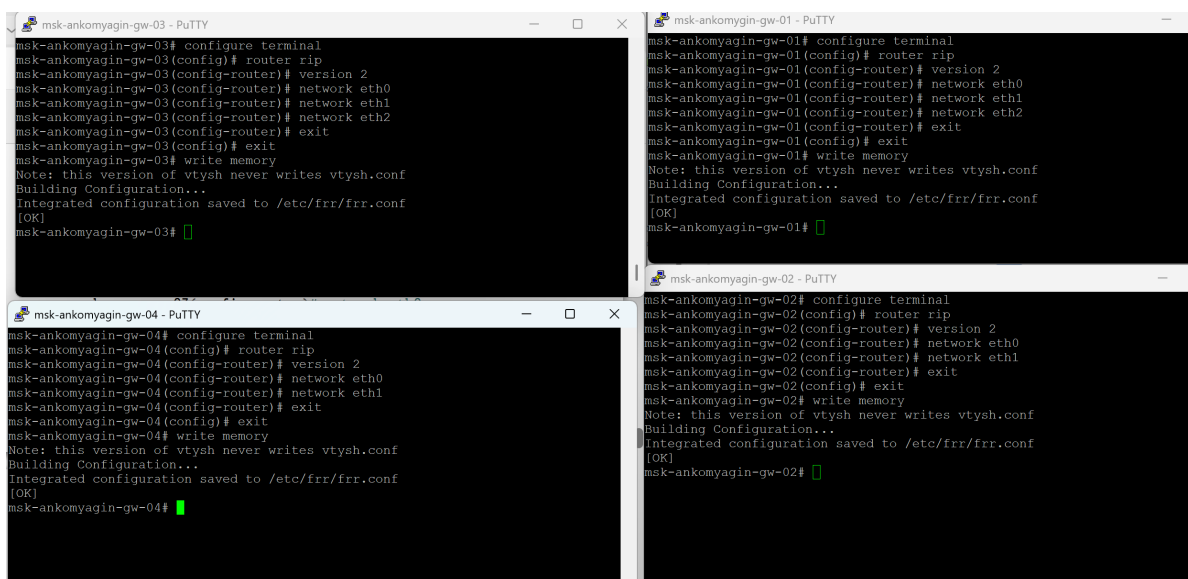


Рис. 3.10: Настройка IPv4 на PC1 и проверка show ip

```
msk-ankomyagin-gw-01 - PuTTY
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/3] via 10.0.4.1, eth2, weight 1, 00:01:32
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:01:32
R>* 10.0.11.0/24 [120/3] via 10.0.4.1, eth2, weight 1, 00:01:32
msk-ankomyagin-gw-01# ip rip
% Unknown command: ip rip
msk-ankomyagin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          3 10.0.4.1           0 02:51
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:51
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          3 10.0.4.1           0 02:51
msk-ankomyagin-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 8 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth1           2     2
    eth2           2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.4.1      0           0          120      00:00:13
  Distance: (default is 120)
msk-ankomyagin-gw-01#
```

Рис. 3.11: Настройка IPv4 на PC2 и проверка show ip

Оконечным устройствам назначены IPv6-адреса командами VPCS в соответствии с табл. 8.2 (PC1: 2001:10::a/64; PC2: 2001:11::a/64).

```
PC1> ip 2001:10::a/64
```

```
PC1> save
```

```
PC1> show ipv6
```

```
PC2> ip 2001:11::a/64
```

```
PC2> save
```

```
PC2> show ipv6
```

```
PC1-ankomyagin - PuTTY
VPCS> ping 10.0.11.10

10.0.11.10 icmp_seq=1 timeout
10.0.11.10 icmp_seq=2 timeout
10.0.11.10 icmp_seq=3 timeout
10.0.11.10 icmp_seq=4 timeout
10.0.11.10 icmp_seq=5 timeout

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=16.234 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=17.448 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=14.628 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=15.369 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=18.476 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    8.401 ms  3.170 ms  3.355 ms
 2  10.0.4.1    12.157 ms  9.484 ms  6.503 ms
 3  10.0.3.1    15.682 ms 15.300 ms 13.443 ms
 4  10.0.11.10  14.339 ms 12.500 ms 14.462 ms

VPCS> █

msk-ankomyagin-gw-01 - PuTTY
msk-ankomyagin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          3 10.0.4.1           0 02:36
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:36
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          3 10.0.4.1           0 02:36
msk-ankomyagin-gw-01# █
```

Рис. 3.12: Настройка IPv6 на PC1 и проверка show ipv6

```
msk-ankomyagin-gw-02 - PuTTY
msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# sshutdown
% Unknown command: sshutdown
msk-ankomyagin-gw-02(config-if)# shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# exit

msk-ankomyagin-gw-01 - PuTTY
msk-ankomyagin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.4.1          3 10.0.4.1           0 02:40
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:40
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          3 10.0.4.1           0 02:40
msk-ankomyagin-gw-01#

PC1-ankomyagin - PuTTY
VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1 10.0.10.1   8.401 ms  3.170 ms  3.355 ms
 2 10.0.4.1   12.157 ms 9.484 ms  6.503 ms
 3 10.0.3.1   15.682 ms 15.300 ms 13.443 ms
 4 10.0.11.10 14.339 ms 12.500 ms 14.462 ms

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=14.591 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=10.988 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=14.154 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=15.569 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=17.560 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1 10.0.10.1   3.389 ms  1.181 ms  1.707 ms
 2 10.0.4.1    6.839 ms  3.823 ms  6.301 ms
 3 10.0.3.1   10.092 ms 9.512 ms  8.033 ms
 4 10.0.11.10 12.289 ms 17.517 ms 10.660 ms

VPCS>
```

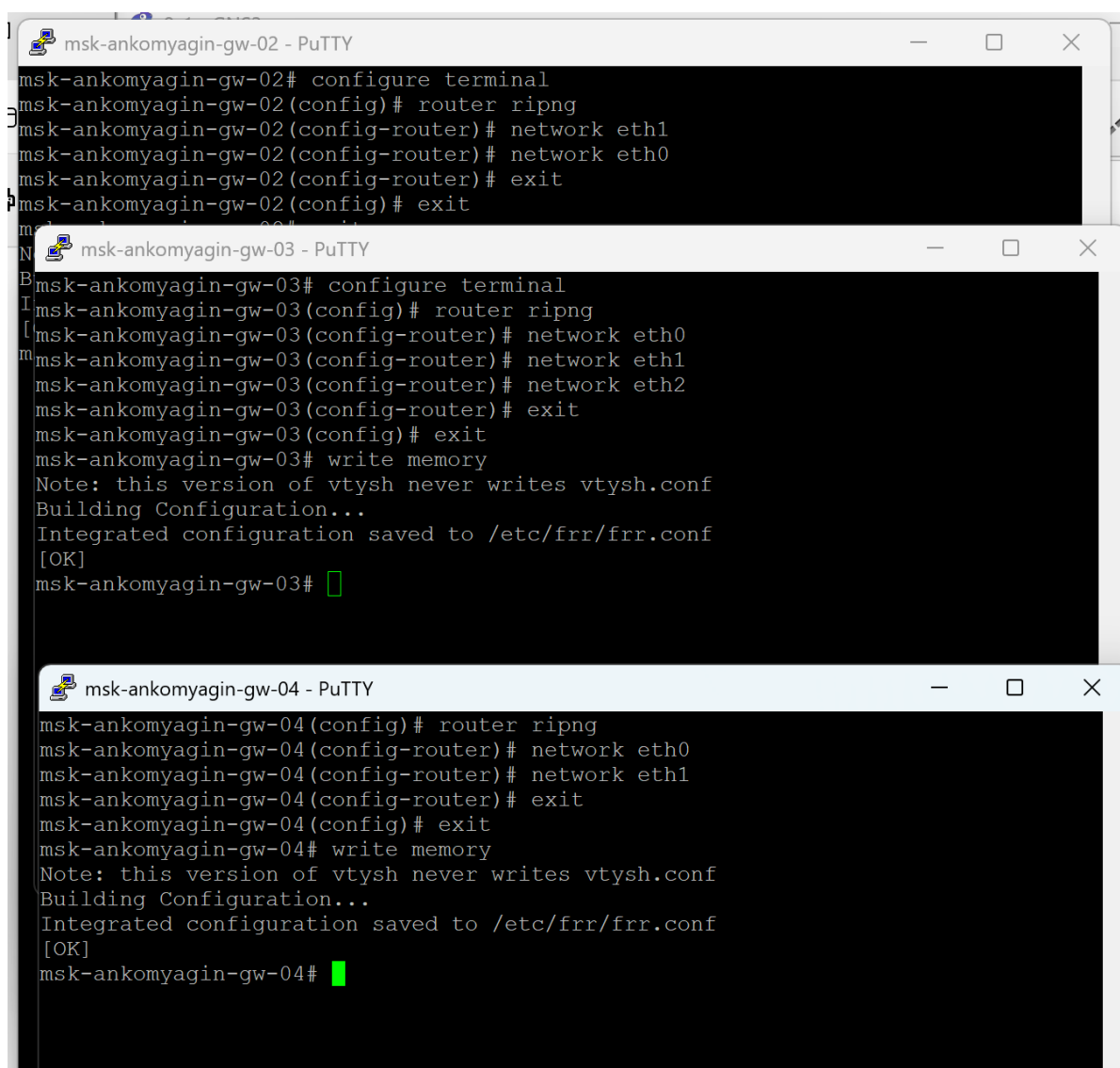
Рис. 3.13: Настройка IPv6 на PC2 и проверка show ipv6

3.3.2 Настройка адресов на маршрутизаторах (FRR)

На маршрутизаторах msk-ankomyagin-gw-01...msk-ankomyagin-gw-04 настроены IPv4-адреса интерфейсов согласно табл. 8.2 и сохранена конфигурация (write memory).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4867, seq=1/256, ttl=61 (reply in 4)
2	0.000629	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.11.1? Tell 10.0.11.10
3	0.002456	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	10.0.11.1 is at 0c:57:e5:b5:00:00
4	0.003971	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4867, seq=1/256, ttl=64 (request in 1)
5	1.017646	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4967, seq=2/512, ttl=61 (reply in 6)
6	1.019047	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4967, seq=2/512, ttl=64 (request in 5)
7	2.033034	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4a67, seq=3/768, ttl=61 (reply in 8)
8	2.033997	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4a67, seq=3/768, ttl=64 (request in 7)
9	3.050207	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4b67, seq=4/1024, ttl=61 (reply in 10)
10	3.050680	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4b67, seq=4/1024, ttl=64 (request in 9)
11	4.064079	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4c67, seq=5/1280, ttl=61 (reply in 12)
12	4.064271	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4c67, seq=5/1280, ttl=64 (request in 11)
13	5.008728	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	Who has 10.0.11.10? Tell 10.0.11.1
14	5.009112	Private_66:68:01	0c:57:e5:b5:00:00	ARP	60	10.0.11.10 is at 00:50:79:66:68:01
15	13.739274	10.0.11.1	224.0.0.9	RIPv2	146	Response

Рис. 3.14: Настройка IPv4 на msk-ankomyagin-gw-01 (eth0/eth1/eth2)



```
msk-ankomyagin-gw-02 - PuTTY
msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# router ripng
msk-ankomyagin-gw-02(config-router)# network eth1
msk-ankomyagin-gw-02(config-router)# network eth0
msk-ankomyagin-gw-02(config-router)# exit
msk-ankomyagin-gw-02(config)# exit

msk-ankomyagin-gw-03 - PuTTY
msk-ankomyagin-gw-03# configure terminal
msk-ankomyagin-gw-03(config)# router ripng
msk-ankomyagin-gw-03(config-router)# network eth0
msk-ankomyagin-gw-03(config-router)# network eth1
msk-ankomyagin-gw-03(config-router)# network eth2
msk-ankomyagin-gw-03(config-router)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03#

msk-ankomyagin-gw-04 - PuTTY
msk-ankomyagin-gw-04(config)# router ripng
msk-ankomyagin-gw-04(config-router)# network eth0
msk-ankomyagin-gw-04(config-router)# network eth1
msk-ankomyagin-gw-04(config-router)# exit
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04#
```

Рис. 3.15: Настройка IPv4 на msk-ankomyagin-gw-02 (eth0/eth1)


```
PC1-ankomyagin - PuTTY

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=29.912 ms
2001:11::a icmp6_seq=2 ttl=58 time=22.292 ms
2001:11::a icmp6_seq=3 ttl=58 time=20.132 ms
2001:11::a icmp6_seq=4 ttl=58 time=20.142 ms
2001:11::a icmp6_seq=5 ttl=58 time=16.571 ms

VPCS> trace 2001:11::a

Trace to 2001:11::a, 64 hops max
 1 2001:10::1    9.558 ms  2.065 ms  1.852 ms
 2 2001:1::2    20.331 ms 9.236 ms  8.694 ms
 3 2001:2::2    12.896 ms 12.064 ms 15.563 ms
 4 2001:11::a   13.465 ms 17.022 ms 19.011 ms

VPCS>

msk-ankomyagin-gw-01 - PuTTY

msk-ankomyagin-gw-01(config-router)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

   Network      Next Hop          Via      Metric Tag Time
C(i) 2001:1::/64      ::              self        1    0
R(n) 2001:2::/64      fe80::ef1:5ff:fe00:0 eth1         2    0 02:30
R(n) 2001:3::/64      fe80::e00:91ff:fe54:1 eth2         2    0 02:50
C(i) 2001:4::/64      ::              self        1    0
C(i) 2001:10::/64     ::              self        1    0
R(n) 2001:11::/64     fe80::ef1:5ff:fe00:0 eth1         3    0 02:50
msk-ankomyagin-gw-01#
```

Рис. 3.16: Настройка IPv4 на msk-ankomyagin-gw-03 (eth0/eth1/eth2)

```

PC1-ankomyagin - PuTTY
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=15.217 ms
2001:11::a icmp6_seq=2 ttl=58 time=16.405 ms
2001:11::a icmp6_seq=3 ttl=58 time=19.472 ms
2001:11::a icmp6_seq=4 ttl=58 time=16.335 ms
2001:11::a icmp6_seq=5 ttl=58 time=16.124 ms
VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1    3.397 ms   3.916 ms   4.130 ms
 2 2001:1::2     7.837 ms   8.219 ms   5.065 ms
 3 2001:2::2     19.078 ms  14.993 ms   7.681 ms
 4 2001:11::a    13.131 ms  13.768 ms  16.898 ms
VPCS>

msk-ankomyagin-gw-01 - PuTTY
fe80::ef1:5ff:fe00:0    eth1    3    0    02:53
C(i) 2001:4::/64
::                      self    1    0
C(i) 2001:10::/64
::                      self    1    0
R(n) 2001:11::/64
fe80::ef1:5ff:fe00:0    eth1    3    0    02:53
msk-ankomyagin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64
::           self      1        0
R(n) 2001:2::/64
fe80::ef1:5ff:fe00:0    eth1      2        0    02:50
R(n) 2001:3::/64
fe80::ef1:5ff:fe00:0    eth1      3        0    02:50
C(i) 2001:4::/64
::           self      1        0
C(i) 2001:10::/64
::           self      1        0
R(n) 2001:11::/64
fe80::ef1:5ff:fe00:0    eth1      3        0    02:50
msk-ankomyagin-gw-01# show ipv6 ripng

```

Рис. 3.17: Настройка IPv4 на msk-ankomyagin-gw-04 (eth0/eth1)

На маршрутизаторах включена маршрутизация IPv6 (ipv6 forwarding), назначены IPv6-адреса интерфейсам согласно табл. 8.2, а на интерфейсах к хостам разрешена рассылка RA (по ipv6 nd suppress-ra) и задан префикс для ND.

No.	Time	Source	Destination	Protocol	Length	Info
120	644.889312	10.0.11.1	224.0.0.9	RIPv2	106	Response
121	644.928994	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xca69, seq=4, hop limit=61 (reply in 122)
122	644.929289	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xca69, seq=4, hop limit=61 (request in 121)
123	645.948447	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xca69, seq=5, hop limit=61 (reply in 124)
124	645.949001	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xca69, seq=5, hop limit=61 (request in 123)
125	654.903159	10.0.11.1	224.0.0.9	RIPv2	146	Response
126	657.513268	fe80::e57:e5ff:feb5...	ff02::9	RIPng	86	Command Response, Version 1
127	658.917486	10.0.11.1	224.0.0.9	RIPv2	106	Response
128	661.487918	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
129	661.488403	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
130	661.503313	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
131	661.503508	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
132	661.522954	2001:10::a	2001:11::a	UDP	126	27792 → 27793 Len=64
133	661.524996	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
134	666.526981	fe80::e57:e5ff:feb5...	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
135	667.568915	fe80::e57:e5ff:feb5...	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
136	668.607606	fe80::e57:e5ff:feb5...	2001:11::a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
137	669.000436	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
138	679.911372	10.0.11.1	224.0.0.9	RIPv2	146	Response
139	700.008266	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
140	712.917352	10.0.11.1	224.0.0.9	RIPv2	146	Response
141	725.017226	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
142	737.740436	fe80::e57:e5ff:feb5...	ff02::1:ff00:a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
143	737.742183	2001:11::a	fe80::e57:e5ff:feb5...	ICMPv6	86	Neighbor Advertisement 2001:11::a (sol, ovr) is at 00:50:79:66:68:01
144	737.747028	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=1, hop limit=61 (reply in 145)
145	737.747983	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=1, hop limit=61 (request in 144)
146	738.763907	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=2, hop limit=61 (reply in 147)
147	738.765250	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=2, hop limit=61 (request in 146)
148	739.783592	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=3, hop limit=61 (reply in 149)
149	739.784009	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=3, hop limit=61 (request in 148)
150	740.635221	10.0.11.1	224.0.0.9	RIPv2	146	Response
151	740.803883	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=4, hop limit=61 (reply in 152)
152	740.804272	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=4, hop limit=61 (request in 151)
153	741.822632	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x296a, seq=5, hop limit=61 (reply in 154)
154	741.824262	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x296a, seq=5, hop limit=61 (request in 153)

Рис. 3.18: Настройка IPv6 на msk-ankomyagin-gw-01 и параметры ND/RA на eth0

msk-ankomyagin-gw-01 - PuTTY

```

msk-ankomyagin-gw-01# configure terminal
msk-ankomyagin-gw-01(config)# router ospf
msk-ankomyagin-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-ankomyagin-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-ankomyagin-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-ankomyagin-gw-01(config-router)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01#

```

msk-ankomyagin-gw-02 - PuTTY

```

msk-ankomyagin-gw-02(config)# router ospf
msk-ankomyagin-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-ankomyagin-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-ankomyagin-gw-02(config-router)# exit
msk-ankomyagin-gw-02(config)# exit
msk-ankomyagin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-02#

```

msk-ankomyagin-gw-03 - PuTTY

```

msk-ankomyagin-gw-03# configure
msk-ankomyagin-gw-03(config)# router ospf
msk-ankomyagin-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-ankomyagin-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-ankomyagin-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-ankomyagin-gw-03(config-router)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03#

```

msk-ankomyagin-gw-04 - PuTTY

```

msk-ankomyagin-gw-04(config)# router ospf
msk-ankomyagin-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-ankomyagin-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-ankomyagin-gw-04(config-router)# exit
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04#

```

Рис. 3.19: Настройка IPv6 на msk-ankomyagin-gw-02

```
PC1-ankomyagin - PuTTY
2001:11::a icmp6_seq=5 ttl=58 time=22.738 ms
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=25.009 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=18.087 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=12.909 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=23.663 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=21.834 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    4.865 ms  0.797 ms  2.716 ms
 2  10.0.4.1     11.404 ms 5.913 ms 12.336 ms
 3  10.0.3.1     17.774 ms 14.900 ms 12.250 ms
 4  10.0.11.10   17.882 ms 14.910 ms 13.984 ms
VPCS> █

msk-ankomyagin-gw-01 - PuTTY
msk-ankomyagin-gw-01# show ip ospf neighbor

Neighbor ID      Pri State           Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.1.2         1 Init/DROther     3m51s            38.302s 10.0.1.2      et
h1:10.0.1.1
10.0.4.1         1 Full/Backup     1m42s            34.189s 10.0.4.1      et
h2:10.0.4.2
0                0                0

msk-ankomyagin-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24      [300] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.3.0/24      [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.4.1, eth2

===== OSPF router routing table =====
===== OSPF external routing table =====

msk-ankomyagin-gw-01# █
```

Рис. 3.20: Настройка IPv6 на msk-ankomyagin-gw-03 и параметры ND/RA на eth0

```
msk-ankomyagin-gw-02 - PuTTY
[OK]
msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# shutdown
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)#
msk-ankomyagin-gw-02(config)#
msk-ankomyagin-gw-02(config)#

msk-ankomyagin-gw-01 - PuTTY
===== OSPF external routing table =====

msk-ankomyagin-gw-01# show ip ospf neighbor

Neighbor ID      Pri State           Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.1.2          1 Init/DROther     4m29s            30.666s 10.0.1.2      et
hl:10.0.1.1
10.0.4.1          1 Full/Backup     14m17s           39.665s 10.0.4.1      et
h2:10.0.4.2
0 0 0

msk-ankomyagin-gw-01# show ip ospf route
===== OSPF network routing table =====
N 10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N 10.0.2.0/24      [300] area: 0.0.0.0
    via 10.0.4.1, eth2
N 10.0.3.0/24      [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N 10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N 10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N 10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.4.1, eth2

===== OSPF router routing table =====
===== OSPF external routing table =====
```

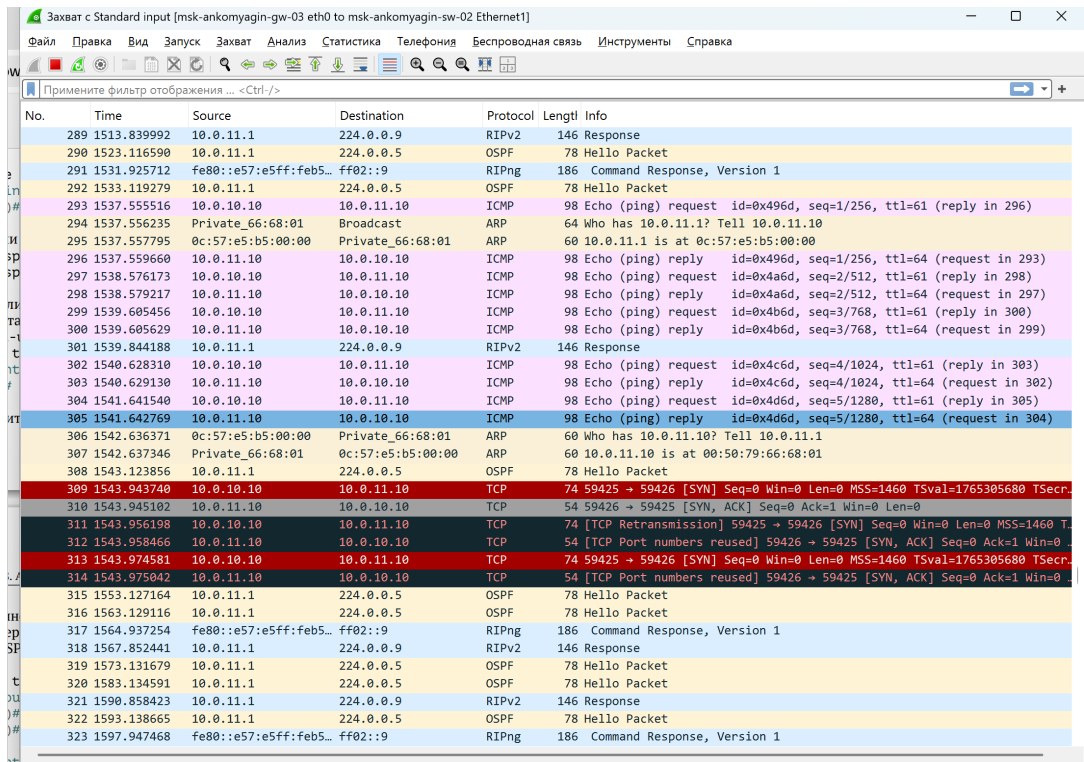
Рис. 3.21: Настройка IPv6 на msk-ankomyagin-gw-04

3.4 Динамическая маршрутизация RIP/RIPng

3.4.1 RIP для IPv4

RIP — дистанционно-векторный протокол; в RIPv2 обновления передаются на multicast-адрес 224.0.0.9, а метрика определяется числом транзитных переходов.

На всех маршрутизаторах включён RIP версии 2 и добавлены сети через перечисление интерфейсов (network ethX) согласно методике задания.



No.	Time	Source	Destination	Protocol	Length	Info
289	1513.839992	10.0.11.1	224.0.0.9	RIPv2	146	Response
290	1523.116590	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
291	1531.925712	fe80::e57:e5ff:feb5... ff02::9		RIPng	186	Command Response, Version 1
292	1533.119279	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
293	1537.555516	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x496d, seq=1/256, ttl=61 (reply in 296)
294	1537.556235	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.11.1? Tell 10.0.11.10
295	1537.557795	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	10.0.11.1 is at 0c:57:e5:b5:00:00
296	1537.559660	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x496d, seq=1/256, ttl=64 (request in 293)
297	1538.576173	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4a6d, seq=2/512, ttl=61 (reply in 298)
298	1538.579217	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4a6d, seq=2/512, ttl=64 (request in 297)
299	1539.605456	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4b6d, seq=3/768, ttl=61 (reply in 300)
300	1539.605629	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4b6d, seq=3/768, ttl=64 (request in 299)
301	1539.844188	10.0.11.1	224.0.0.9	RIPv2	146	Response
302	1540.628310	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4c6d, seq=4/1024, ttl=61 (reply in 303)
303	1540.629130	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4c6d, seq=4/1024, ttl=64 (request in 302)
304	1541.641540	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x4d6d, seq=5/1280, ttl=61 (reply in 305)
305	1541.642769	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x4d6d, seq=5/1280, ttl=64 (request in 304)
306	1542.636371	0c:57:e5:b5:00:00	Private_66:68:01	ARP	60	Who has 10.0.11.10? Tell 10.0.11.1
307	1542.637346	Private_66:68:01	0c:57:e5:b5:00:00	ARP	60	10.0.11.10 is at 00:50:79:66:68:01
308	1543.123856	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
309	1543.943740	10.0.10.10	10.0.11.10	TCP	74	59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765305680 TSecr...
310	1543.945102	10.0.11.10	10.0.10.10	TCP	54	59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
311	1543.956198	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 T...
312	1543.958466	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0
313	1543.974581	10.0.10.10	10.0.11.10	TCP	74	59425 → 59426 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765305680 TSecr...
314	1543.975042	10.0.11.10	10.0.10.10	TCP	54	[TCP Port numbers reused] 59426 → 59425 [SYN, ACK] Seq=0 Ack=1 Win=0
315	1553.127164	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
316	1563.129116	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
317	1564.937254	fe80::e57:e5ff:feb5... ff02::9		RIPng	186	Command Response, Version 1
318	1567.852441	10.0.11.1	224.0.0.9	RIPv2	146	Response
319	1573.131679	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
320	1583.134591	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
321	1590.858423	10.0.11.1	224.0.0.9	RIPv2	146	Response
322	1593.138665	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
323	1597.947468	fe80::e57:e5ff:feb5... ff02::9		RIPng	186	Command Response, Version 1

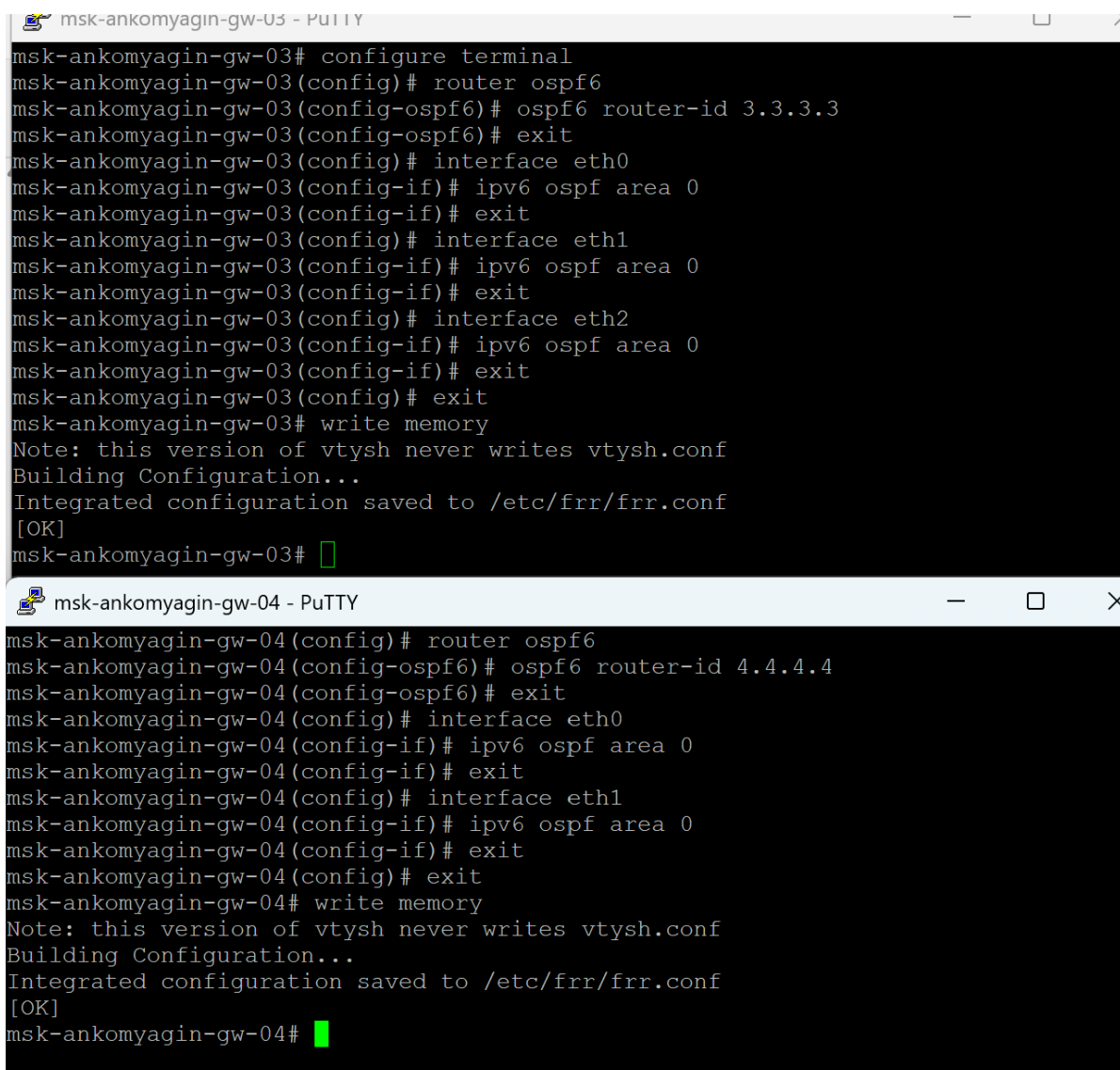
Рис. 3.22: Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-01

The image displays two PuTTY terminal windows. The top window, titled 'msk-ankomyagin-gw-01 - PuTTY', shows the configuration of msk-ankomyagin-gw-01. The user enters 'configure terminal', then 'router ospf6', followed by 'ospf router-id 1.1.1.1'. They then exit to the configuration mode and configure three interfaces: 'eth0', 'eth1', and 'eth2', each with 'ipv6 ospf6 area 0'. After exiting the interface configurations, they use 'write memory' to save the configuration. A note indicates that this version of vtysh does not write to vtysh.conf, and the integrated configuration is saved to /etc/frr/frr.conf. The bottom window, titled 'msk-ankomyagin-gw-02 - PuTTY', shows the configuration of msk-ankomyagin-gw-02. The user enters 'router ospf6', followed by 'ospf6 router-id 2.2.2.2'. They then exit to the configuration mode and configure two interfaces: 'eth0' and 'eth1', each with 'ospf6 area 0'. After exiting the interface configurations, they use 'write memory' to save the configuration. A note indicates that this version of vtysh does not write to vtysh.conf, and the integrated configuration is saved to /etc/frr/frr.conf.

```
msk-ankomyagin-gw-01# configure terminal
msk-ankomyagin-gw-01(config)# router ospf6
msk-ankomyagin-gw-01(config-ospf6)# ospf router-id 1.1.1.1
msk-ankomyagin-gw-01(config-ospf6)# exit
msk-ankomyagin-gw-01(config)# interface eth0
msk-ankomyagin-gw-01(config-if)# ipv6 ospf6 area 0
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth1
msk-ankomyagin-gw-01(config-if)# ipv6 ospf6 area 0
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# interface eth2
msk-ankomyagin-gw-01(config-if)# ipv6 ospf6 area 0
msk-ankomyagin-gw-01(config-if)# exit
msk-ankomyagin-gw-01(config)# exit
msk-ankomyagin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-01#

msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# router ospf6
msk-ankomyagin-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-ankomyagin-gw-02(config-ospf6)# exit
msk-ankomyagin-gw-02(config)# interface eth0
% Unknown command: interface eth0
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# ipv6 ospf6 area 0
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# interface eth1
msk-ankomyagin-gw-02(config-if)# ipv6 ospf6 area 0
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# exit
msk-ankomyagin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-02#
```

Рис. 3.23: Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-02



The image displays two terminal windows from a PuTTY application. The top window, titled 'msk-ankomyagin-gw-03 - PuTTY', shows the configuration of OSPF6 on router msk-ankomyagin-gw-03. The user enters 'configure terminal', then 'router ospf6', followed by 'ospf6 router-id 3.3.3.3'. They then exit the router configuration and enter 'interface eth0', 'ipv6 ospf area 0', and 'exit'. This process is repeated for 'interface eth1' and 'interface eth2'. Finally, they exit the configuration mode and enter 'write memory'. A message states: 'Note: this version of vtysh never writes vtysh.conf Building Configuration... Integrated configuration saved to /etc/frr/frr.conf [OK]'. The prompt returns to 'msk-ankomyagin-gw-03#'. The bottom window, titled 'msk-ankomyagin-gw-04 - PuTTY', shows the same configuration steps for router msk-ankomyagin-gw-04, but with the router ID set to '4.4.4.4'. It also shows the 'write memory' command and the same confirmation message. The prompt returns to 'msk-ankomyagin-gw-04#'.

```
msk-ankomyagin-gw-03# configure terminal
msk-ankomyagin-gw-03(config)# router ospf6
msk-ankomyagin-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-ankomyagin-gw-03(config-ospf6)# exit
msk-ankomyagin-gw-03(config)# interface eth0
msk-ankomyagin-gw-03(config-if)# ipv6 ospf area 0
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ipv6 ospf area 0
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth2
msk-ankomyagin-gw-03(config-if)# ipv6 ospf area 0
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03#

msk-ankomyagin-gw-04# configure terminal
msk-ankomyagin-gw-04(config)# router ospf6
msk-ankomyagin-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-ankomyagin-gw-04(config-ospf6)# exit
msk-ankomyagin-gw-04(config)# interface eth0
msk-ankomyagin-gw-04(config-if)# ipv6 ospf area 0
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ipv6 ospf area 0
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-04#
```

Рис. 3.24: Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-03


```
PC1-ankomyagin - PuTTY
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=24.150 ms
2001:11::a icmp6_seq=2 ttl=58 time=10.587 ms
2001:11::a icmp6_seq=3 ttl=58 time=16.198 ms
2001:11::a icmp6_seq=4 ttl=58 time=13.529 ms
2001:11::a icmp6_seq=5 ttl=58 time=15.942 ms
VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1  2.148 ms  4.893 ms  2.923 ms
 2 2001:4::1  10.400 ms  9.481 ms  7.196 ms
 3 2001:3::1  14.826 ms  13.292 ms  11.498 ms
 4 2001:11::a  20.667 ms  12.560 ms  14.048 ms
VPCS> █

msk-ankomyagin-gw-01 - PuTTY
msk-ankomyagin-gw-01# show ipv6 ospf neighbor
Neighbor ID      Pri   DeadTime      State/IfState      Duration I/F[State]
4.4.4.4          1     00:00:35      Full/BDR           00:01:26 eth2[DR]
msk-ankomyagin-gw-01# show ipv6 ospf route
*N IA 2001:1::/64      ::              eth1 00:01:31
*N IA 2001:2::/64      fe80::e00:91ff:fe54:1 eth2 00:01:31
*N IA 2001:3::/64      fe80::e00:91ff:fe54:1 eth2 00:01:31
*N IA 2001:4::/64      ::              eth2 00:01:31
*N IA 2001:10::/64     ::              eth0 00:01:31
*N IA 2001:11::/64     fe80::e00:91ff:fe54:1 eth2 00:01:31
msk-ankomyagin-gw-01# █
```

Рис. 3.25: Конфигурация RIP (RIPv2) на msk-ankomyagin-gw-04

Корректность настройки проверена командами `show ip route rip`, `show ip rip` и `show ip rip status` на маршрутизаторах.

```
PC1-ankomyagin - PuTTY
VPCS> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=13.944 ms
2001:11::a icmp6_seq=2 ttl=58 time=17.307 ms
2001:11::a icmp6_seq=3 ttl=58 time=14.111 ms
2001:11::a icmp6_seq=4 ttl=58 time=15.997 ms
2001:11::a icmp6_seq=5 ttl=58 time=17.209 ms
VPCS> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1 2.894 ms 2.377 ms 3.496 ms
 2 2001:1::2 5.783 ms 5.608 ms 7.014 ms
 3 2001:2::2 11.273 ms 12.382 ms 11.761 ms
 4 2001:11::a 14.157 ms 14.023 ms 19.565 ms
VPCS> █
Note: this version of which never times out, which goes

msk-ankomyagin-gw-01 - PuTTY
*N IA 2001:4::/64 :: eth2 00:01:31
*N IA 2001:10::/64 :: eth0 00:01:31
*N IA 2001:11::/64 fe80::e00:91ff:fe54:1 eth2 00:01:31
msk-ankomyagin-gw-01# show ipv6 ospf route
*N IA 2001:1::/64 :: eth1 00:02:40
*N IA 2001:2::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
*N IA 2001:3::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
*N IA 2001:4::/64 :: eth2 00:02:40
*N IA 2001:10::/64 :: eth0 00:02:40
*N IA 2001:11::/64 fe80::e00:91ff:fe54:1 eth2 00:02:40
msk-ankomyagin-gw-01# show ipv6 ospf route
*N IA 2001:1::/64 :: eth1 00:00:04
*N IA 2001:2::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04
*N IA 2001:3::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04
*N IA 2001:4::/64 :: eth2 00:00:04
*N IA 2001:10::/64 :: eth0 00:00:04
*N IA 2001:11::/64 fe80::ef1:5ff:fe00:0 eth1 00:00:04
msk-ankomyagin-gw-01#
```

Рис. 3.26: Проверка таблицы RIP-маршрутов (show ip route rip)

Связность проверена с PC1 до PC2 командами ping и trace, при этом маршрут следования пакетов фиксируется в результате trace.

Захват с Standard input [msk-ankomyagin-gw-03 eth0 to msk-ankomyagin-sw-02 Ethernet1]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
631	2715.870259	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
632	2723.615368	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
633	2723.754010	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
634	2725.620065	10.0.11.1	224.0.0.9	RIPv2	146	Response
635	2725.872556	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
636	2733.617590	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
637	2735.876013	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
638	2742.761442	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
639	2743.619921	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
640	2745.883240	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
641	2753.622396	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
642	2754.624682	10.0.11.1	224.0.0.9	RIPv2	146	Response
643	2755.887866	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
644	2756.493317	fe80::e57:e5ff:feb5...	ff02::1:ff00:a	ICMPv6	86	Neighbor Solicitation for 2001:11::a from 0c:57:e5:b5:00:00
645	2756.493675	2001:11::a	fe80::e57:e5ff:feb5...	ICMPv6	86	Neighbor Advertisement 2001:11::a (sol, ovr) is at 00:50:79:66:68:01
646	2756.495143	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=1, hop limit=61 (reply in 647)
647	2756.495284	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=1, hop limit=61 (request in 646)
648	2757.513990	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=2, hop limit=61 (reply in 649)
649	2757.514218	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=2, hop limit=61 (request in 648)
650	2758.527000	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=3, hop limit=61 (reply in 651)
651	2758.528332	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=3, hop limit=61 (request in 650)
652	2759.544325	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=4, hop limit=61 (reply in 653)
653	2759.546177	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=4, hop limit=61 (request in 652)
654	2760.560318	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0x0c72, seq=5, hop limit=61 (reply in 655)
655	2760.561233	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x0c72, seq=5, hop limit=61 (request in 654)
656	2761.896870	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
657	2761.898502	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
658	2761.912715	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
659	2761.913875	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
660	2761.930872	2001:10::a	2001:11::a	UDP	126	48177 → 48178 Len=64
661	2761.932546	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)[Malformed Packet]
662	2763.625720	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet
663	2765.893143	fe80::e57:e5ff:feb5...	ff02::5	OSPF	90	Hello Packet
664	2768.765631	fe80::e57:e5ff:feb5...	ff02::9	RIPng	186	Command Response, Version 1
665	2773.629056	10.0.11.1	224.0.0.5	OSPF	78	Hello Packet

Рис. 3.27: Ping PC1 → PC2 (IPv4) после включения RIP

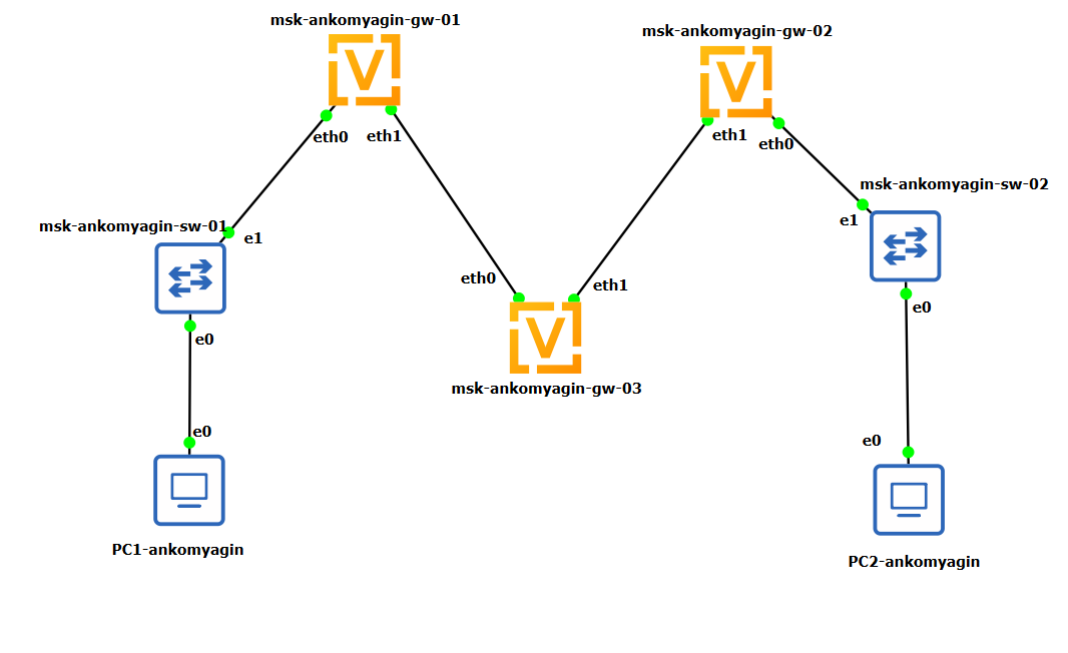


Рис. 3.28: Trace PC1 → PC2 (IPv4) и определение пути

Далее был отключён интерфейс на msk-ankomyagin-gw-02 (shutdown на eth0) для проверки сходимости и переключения на альтернативный путь, после чего интерфейс включён обратно (no shutdown).

The image displays two terminal windows from a PuTTY session. The top window, titled 'PC1-ankomyagin - PuTTY', shows the execution of a startup file, an error message about a long hostname, and the configuration of interface VPCS[1] with IP 1000::a/64. The bottom window, titled 'PC2-ankomyagin - PuTTY', shows similar steps but with IP 1002::a/64. Both windows show the 'show ip' command output, which includes details like MAC address, LPORT, and RHOST:PORT.

```
PC1-ankomyagin - PuTTY
Executing the startup file
::Hostname is too long. (Maximum 12 characters)
VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10012
RHOST:PORT : 127.0.0.1:10013
MTU        : 1500
VPCS>

PC2-ankomyagin - PuTTY
Executing the startup file
Hostname is too long. (Maximum 12 characters)

VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 0.0.0.0/0
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10014
RHOST:PORT : 127.0.0.1:10015
MTU        : 1500
VPCS>
```

Рис. 3.29: Отключение eth0 на msk-ankomyagin-gw-02 для проверки перестроения маршрута

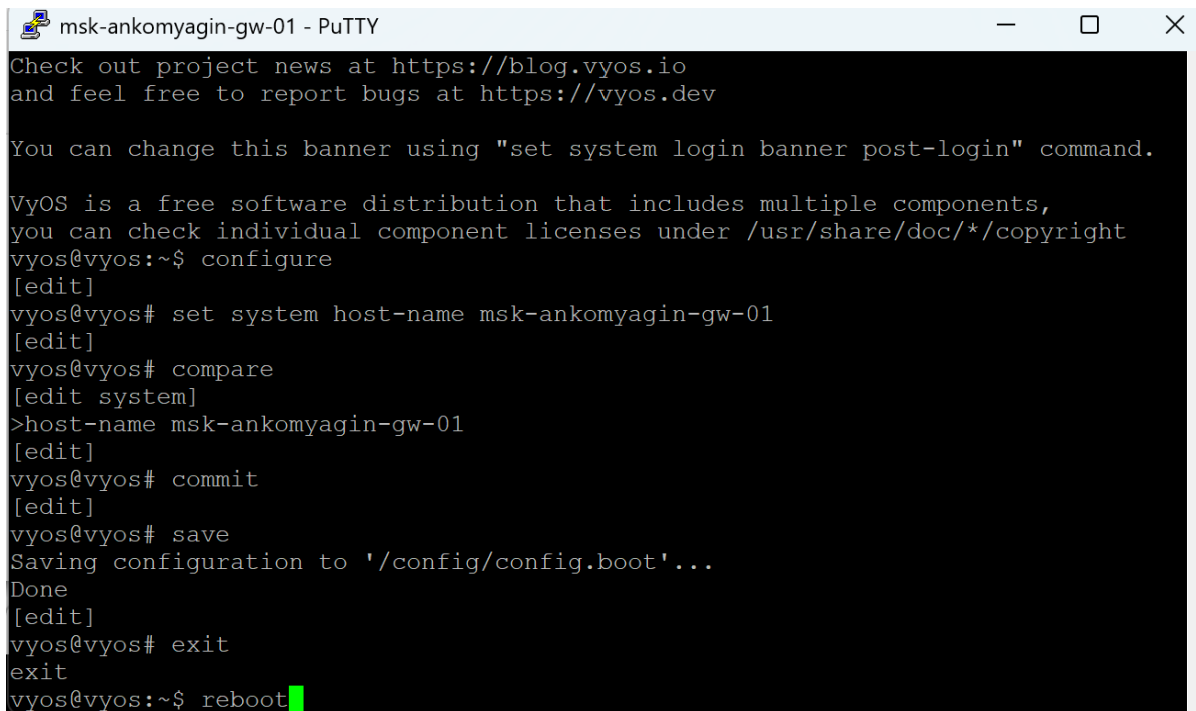


Рис. 3.30: Повторный ping/trace после отключения линка (IPv4, RIP)

В захваченном трафике RIP можно выделить периодические обновления маршрутов (multicast 224.0.0.9), а также изменения объявлений при отказе/восстановлении интерфейса.

```
msk-ankomyagin-gw-01 - PuTTY
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
:/64@msk-ankomyagin-gw-01# set service router-advert interface eth0 prefix 1000:
[edit]
vyos@msk-ankomyagin-gw-01# commit
[edit]
vyos@msk-ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-01#

msk-ankomyagin-gw-02 - PuTTY
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
:/64@msk-ankomyagin-gw-02# set service router-advert interface eth0 prefix 1002:
[edit]
vyos@msk-ankomyagin-gw-02# commit
[edit]
vyos@msk-ankomyagin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-02#

msk-ankomyagin-gw-03 - PuTTY
You can change this banner using "set system login banner post-login" command.
vyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-ankomyagin-gw-03:~$ configure
[edit]
vyos@msk-ankomyagin-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
vyos@msk-ankomyagin-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
```

Рис. 3.31: Захват трафика RIP на линке sw-01—gw-01

```
msk-ankomyagin-gw-01 - PuTTY
vyos@msk-ankomyagin-gw-01:~$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=27.1 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.53 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=5.91 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=3.65 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 24ms
rtt min/avg/max/mdev = 3.645/10.290/27.070/9.721 ms
vyos@msk-ankomyagin-gw-01:~$ ping 20.0.0.2
connect: Network is unreachable
vyos@msk-ankomyagin-gw-01:~$ ping 20.0.0.1
connect: Network is unreachable
vyos@msk-ankomyagin-gw-01:~$
```

Рис. 3.32: Захват трафика RIP на линке sw-02—gw-03

3.4.2 RIPng для IPv6

Для IPv6 используется RIPng, который отправляет обновления на multicast-адрес FF02::9, при этом маршрутизация включается на уровне интерфейса.

На всех маршрутизаторах настроен RIPng, добавлены интерфейсы network ethX по заданию, затем выполнена проверка и тестирование путей.


```
msk-ankomyagin-gw-01 - PuTTY
vyos@msk-ankomyagin-gw-01:~$ configure
[edit]
vyos@msk-ankomyagin-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-01# commit
[edit]
vyos@msk-ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]

msk-ankomyagin-gw-02 - PuTTY
[edit]
vyos@msk-ankomyagin-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-02# commit
[edit]
vyos@msk-ankomyagin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-02# █

msk-ankomyagin-gw-03 - PuTTY
vyos@msk-ankomyagin-gw-03:~$ configure
[edit]
vyos@msk-ankomyagin-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-ankomyagin-gw-03# commit
█
```

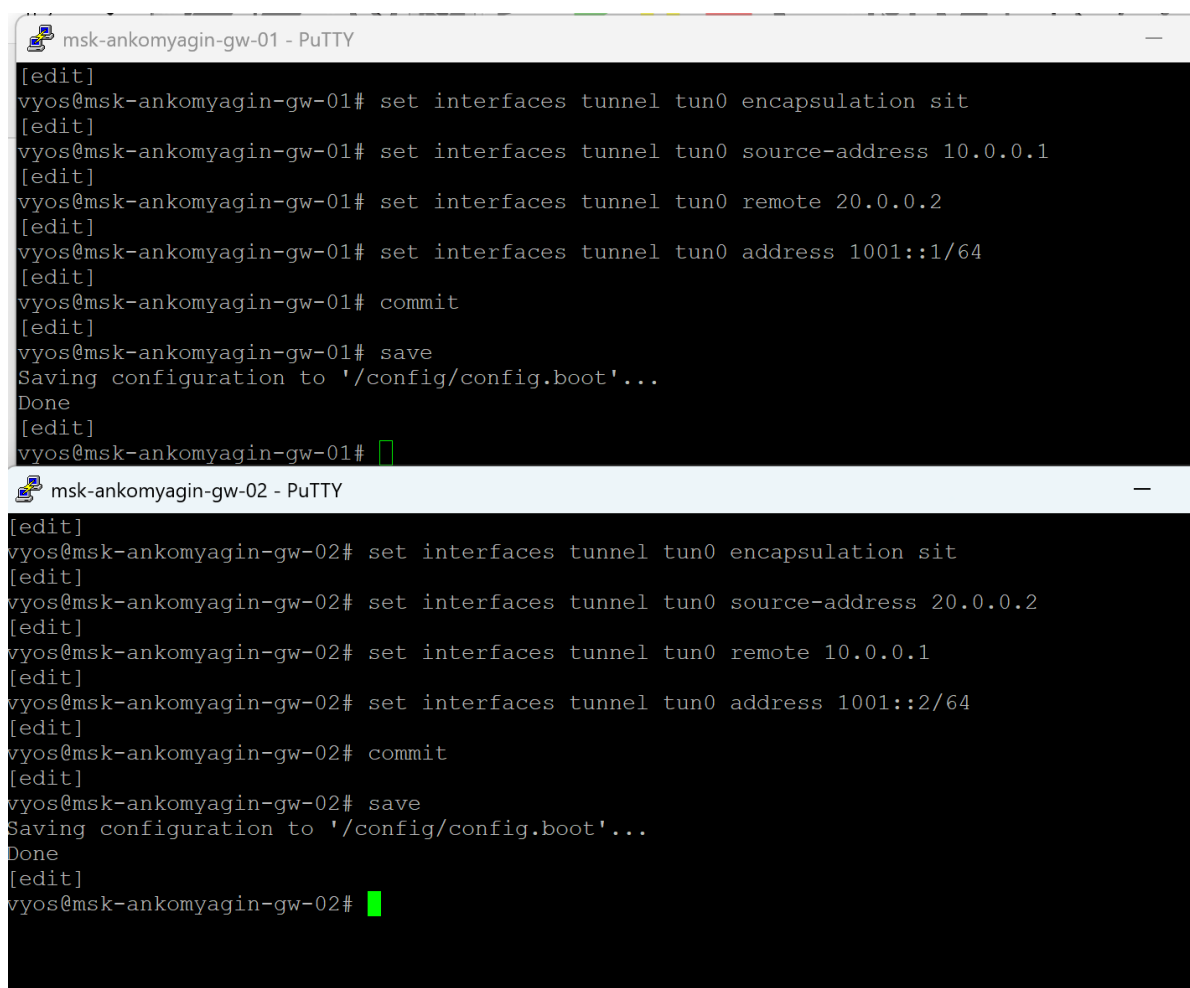
Рис. 3.33: Конфигурация RIPng на msk-ankomyagin-gw-01

```

vyos@msk-ankomyagin-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=18.9 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=10.6 ms
^C
--- 10.0.0.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 10.575/14.723/18.872/4.150 ms
[edit]
vyos@msk-ankomyagin-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=4.92 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=6.16 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=3.27 ms
^C
--- 20.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 12ms
rtt min/avg/max/mdev = 3.266/4.780/6.155/1.186 ms
[edit]
vyos@msk-ankomyagin-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=19.8 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=6.69 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=11.9 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=8.53 ms
^C
--- 20.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 19ms
rtt min/avg/max/mdev = 6.685/11.708/19.752/5.001 ms
[edit]
vyos@msk-ankomyagin-gw-01#

```

Рис. 3.34: Конфигурация RIPng на msk-ankomyagin-gw-02



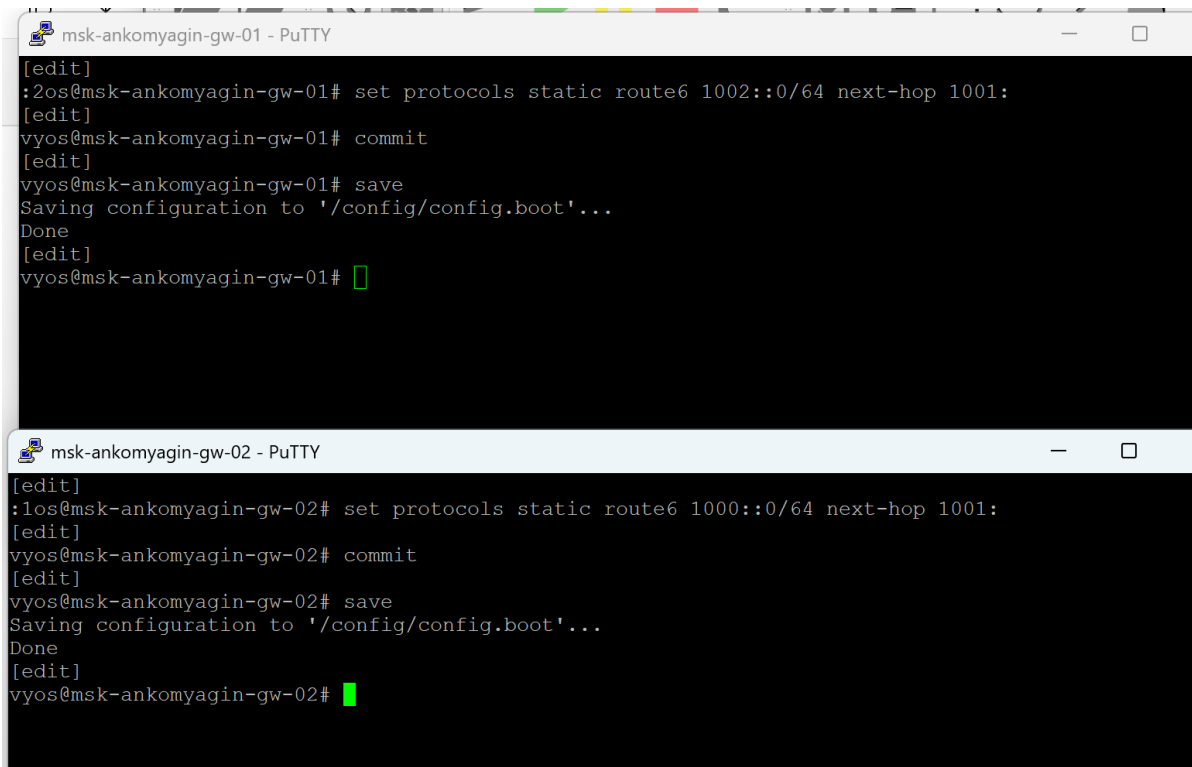
The image shows two terminal windows from PuTTY. The top window is titled 'msk-ankomyagin-gw-01 - PuTTY' and shows the following commands and output:

```
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-ankomyagin-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-ankomyagin-gw-01# commit
[edit]
vyos@msk-ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-01#
```

The bottom window is titled 'msk-ankomyagin-gw-02 - PuTTY' and shows the following commands and output:

```
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-ankomyagin-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-ankomyagin-gw-02# commit
[edit]
vyos@msk-ankomyagin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-02#
```

Рис. 3.35: Конфигурация RIPng на msk-ankomyagin-gw-03



```
msk-ankomyagin-gw-01 - PuTTY
[edit]
:2os@msk-ankomyagin-gw-01# set protocols static route6 1002::0/64 next-hop 1001:
[edit]
vyos@msk-ankomyagin-gw-01# commit
[edit]
vyos@msk-ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-01# █

msk-ankomyagin-gw-02 - PuTTY
[edit]
:1os@msk-ankomyagin-gw-02# set protocols static route6 1000::0/64 next-hop 1001:
[edit]
vyos@msk-ankomyagin-gw-02# commit
[edit]
vyos@msk-ankomyagin-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-ankomyagin-gw-02# █
```

Рис. 3.36: Конфигурация RIPng на msk-ankomyagin-gw-04

Проверка связности для IPv6 выполнена с PC1 до PC2 (ping 2001:11::a, trace 2001:11::a), а также проверены метрики/состояние RIPng на маршрутизаторе (show ipv6 ripng).

```
PC1-ankomyagin - PuTTY
IP/MASK      : 0.0.0.0/0
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:00
LPORT       : 10012
RHOST:PORT   : 127.0.0.1:10013
MTU          : 1500

VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=72.497 ms
1002::a icmp6_seq=2 ttl=60 time=18.305 ms
1002::a icmp6_seq=3 ttl=60 time=20.736 ms
1002::a icmp6_seq=4 ttl=60 time=15.747 ms
1002::a icmp6_seq=5 ttl=60 time=16.208 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1  11.046 ms  2.440 ms  1.106 ms
 2 1001::2  15.843 ms  13.367 ms  9.717 ms
 3 1002::a  22.157 ms  19.700 ms  18.707 ms

PC2-ankomyagin - PuTTY
IP/MASK      : 0.0.0.0/0
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:01
LPORT       : 10014
RHOST:PORT   : 127.0.0.1:10015
MTU          : 1500

VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=14.955 ms
1000::a icmp6_seq=2 ttl=60 time=17.903 ms
1000::a icmp6_seq=3 ttl=60 time=15.293 ms
1000::a icmp6_seq=4 ttl=60 time=13.842 ms
1000::a icmp6_seq=5 ttl=60 time=17.444 ms

VPCS> trace 1000::a

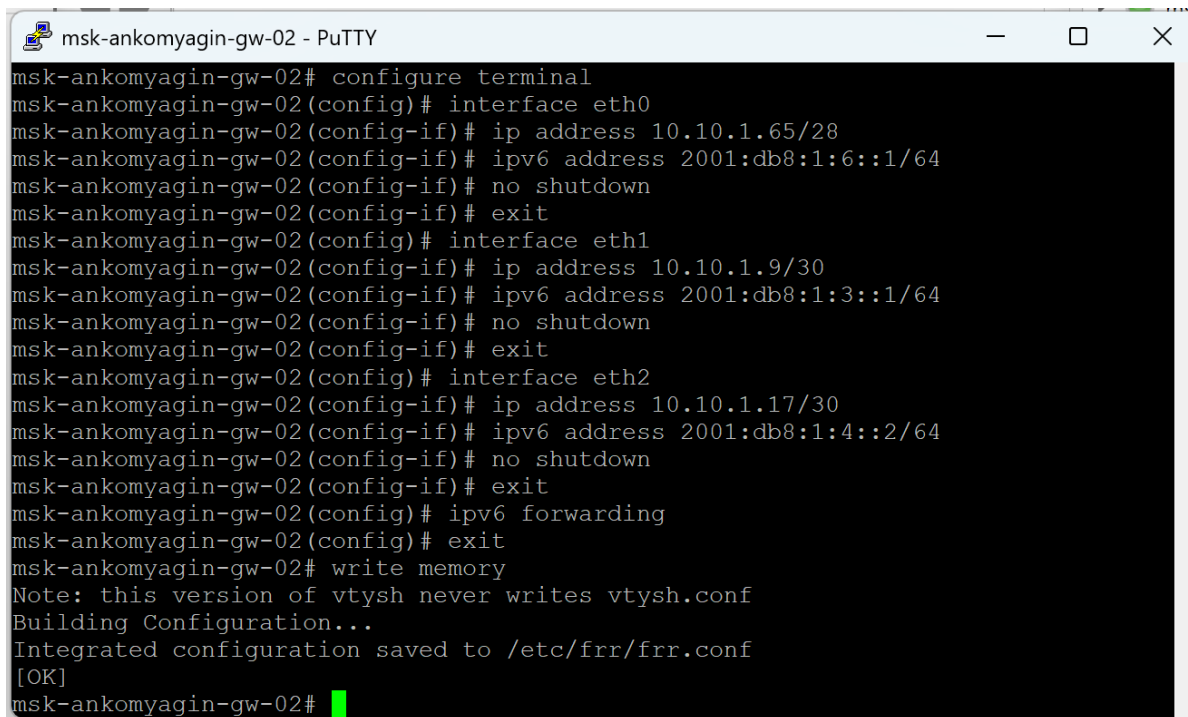
trace to 1000::a, 64 hops max
 1 1002::1  9.160 ms  3.078 ms  2.460 ms
 2 1001::1  11.974 ms  13.318 ms  14.595 ms
 3 1000::a  19.304 ms  17.189 ms  15.866 ms
```

Рис. 3.37: Ping PC1 → PC2 (IPv6) после включения RIPng

Захват с Standard input [msk-ankomyagin-gw-01 eth1 to msk-ankomyagin-gw-03 eth0]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
20	52.752447	1001::2	1000::a	ICMPv6	194	Time Exceeded (Hop limit exceeded in transit)
21	52.758518	1000::a	1002::a	UDP	146	12928 → 12929 Len=64
22	52.762859	1001::2	1000::a	ICMPv6	194	Time Exceeded (Hop limit exceeded in transit)
23	52.773811	1000::a	1002::a	UDP	146	12928 → 12929 Len=64
24	52.784948	1002::a	1000::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
25	52.793252	1000::a	1002::a	UDP	146	12928 → 12929 Len=64
26	52.806901	1002::a	1000::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
27	52.815304	1000::a	1002::a	UDP	146	12928 → 12929 Len=64
28	52.827762	1002::a	1000::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
29	58.068615	10.0.0.2	224.0.0.9	RIPv2	66	Response
30	63.620339	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xca81, seq=1, hop limit=63 (reply in 31)
31	63.625454	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xca81, seq=1, hop limit=61 (request in 30)
32	64.638022	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xca81, seq=2, hop limit=63 (reply in 33)
33	64.643822	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xca81, seq=2, hop limit=61 (request in 32)
34	65.658647	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xca81, seq=3, hop limit=63 (reply in 35)
35	65.664077	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xca81, seq=3, hop limit=61 (request in 34)
36	66.671750	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xca81, seq=4, hop limit=63 (reply in 37)
37	66.676572	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xca81, seq=4, hop limit=61 (request in 36)
38	67.692832	1002::a	1000::a	ICMPv6	138	Echo (ping) request id=0xca81, seq=5, hop limit=63 (reply in 39)
39	67.698436	1000::a	1002::a	ICMPv6	138	Echo (ping) reply id=0xca81, seq=5, hop limit=61 (request in 38)
40	72.971067	0c:aa:f4:be:00:01	0c:aa:f4:be:00:00	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
41	72.971893	0c:aa:f4:be:00:00	0c:77:c2:65:00:01	ARP	60	10.0.0.2 is at 0c:aa:f4:be:00:00
42	74.427448	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
43	74.430410	1001::1	1002::a	ICMPv6	194	Time Exceeded (Hop limit exceeded in transit)
44	74.439303	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
45	74.441865	1001::1	1002::a	ICMPv6	194	Time Exceeded (Hop limit exceeded in transit)
46	74.454629	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
47	74.456765	1001::1	1002::a	ICMPv6	194	Time Exceeded (Hop limit exceeded in transit)
48	74.471519	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
49	74.481491	1000::a	1002::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
50	74.491808	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
51	74.497338	1000::a	1002::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
52	74.508820	1002::a	1000::a	UDP	146	12928 → 12929 Len=64
53	74.515277	1000::a	1002::a	ICMPv6	194	Destination Unreachable (Port unreachable)[Malformed Packet]
54	93.106670	10.0.0.2	224.0.0.9	RIPv2	66	Response

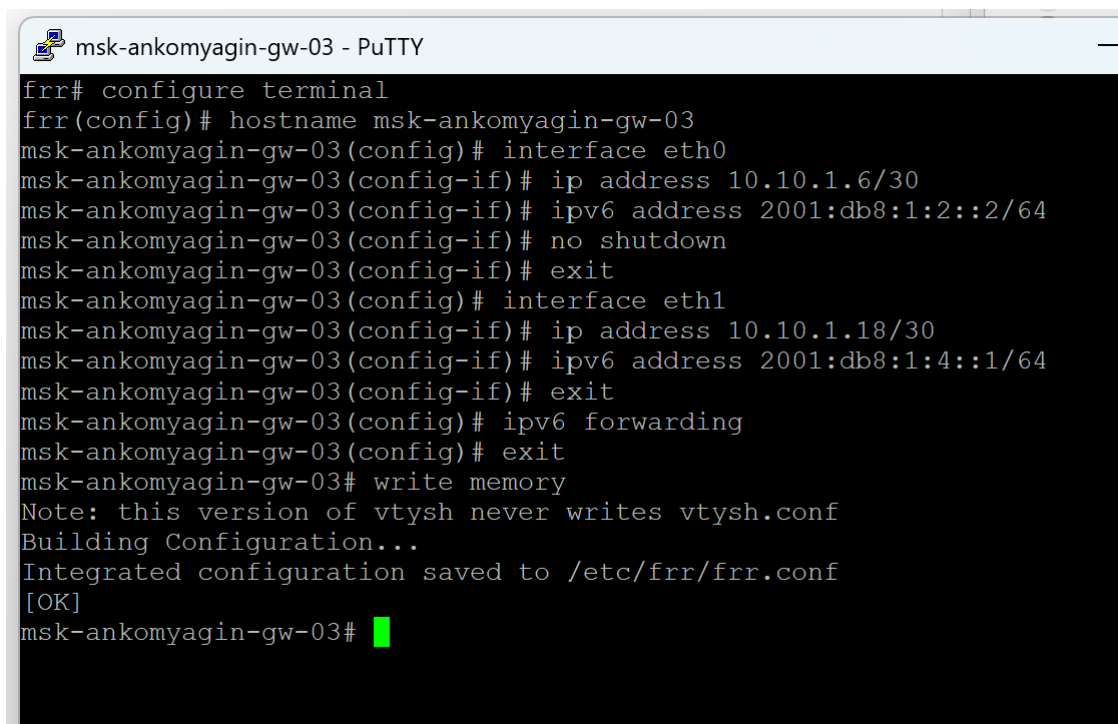
Рис. 3.38: Trace PC1 → PC2 (IPv6) и определение пути

Для проверки отказоустойчивости аналогично отключался интерфейс на msk-ankomyagin-gw-02 (shutdown eth0) и оценивалось перестроение маршрута, после чего интерфейс возвращался в рабочее состояние.



```
msk-ankomyagin-gw-02# configure terminal
msk-ankomyagin-gw-02(config)# interface eth0
msk-ankomyagin-gw-02(config-if)# ip address 10.10.1.65/28
msk-ankomyagin-gw-02(config-if)# ipv6 address 2001:db8:1:6::1/64
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# interface eth1
msk-ankomyagin-gw-02(config-if)# ip address 10.10.1.9/30
msk-ankomyagin-gw-02(config-if)# ipv6 address 2001:db8:1:3::1/64
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# interface eth2
msk-ankomyagin-gw-02(config-if)# ip address 10.10.1.17/30
msk-ankomyagin-gw-02(config-if)# ipv6 address 2001:db8:1:4::2/64
msk-ankomyagin-gw-02(config-if)# no shutdown
msk-ankomyagin-gw-02(config-if)# exit
msk-ankomyagin-gw-02(config)# ipv6 forwarding
msk-ankomyagin-gw-02(config)# exit
msk-ankomyagin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-02#
```

Рис. 3.41: Захват трафика RIPng на линке sw-01—gw-01



```
frr# configure terminal
frr(config)# hostname msk-ankomyagin-gw-03
msk-ankomyagin-gw-03(config)# interface eth0
msk-ankomyagin-gw-03(config-if)# ip address 10.10.1.6/30
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:db8:1:2::2/64
msk-ankomyagin-gw-03(config-if)# no shutdown
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# interface eth1
msk-ankomyagin-gw-03(config-if)# ip address 10.10.1.18/30
msk-ankomyagin-gw-03(config-if)# ipv6 address 2001:db8:1:4::1/64
msk-ankomyagin-gw-03(config-if)# exit
msk-ankomyagin-gw-03(config)# ipv6 forwarding
msk-ankomyagin-gw-03(config)# exit
msk-ankomyagin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-ankomyagin-gw-03#
```

Рис. 3.42: Захват трафика RIPng на линке sw-02—gw-03

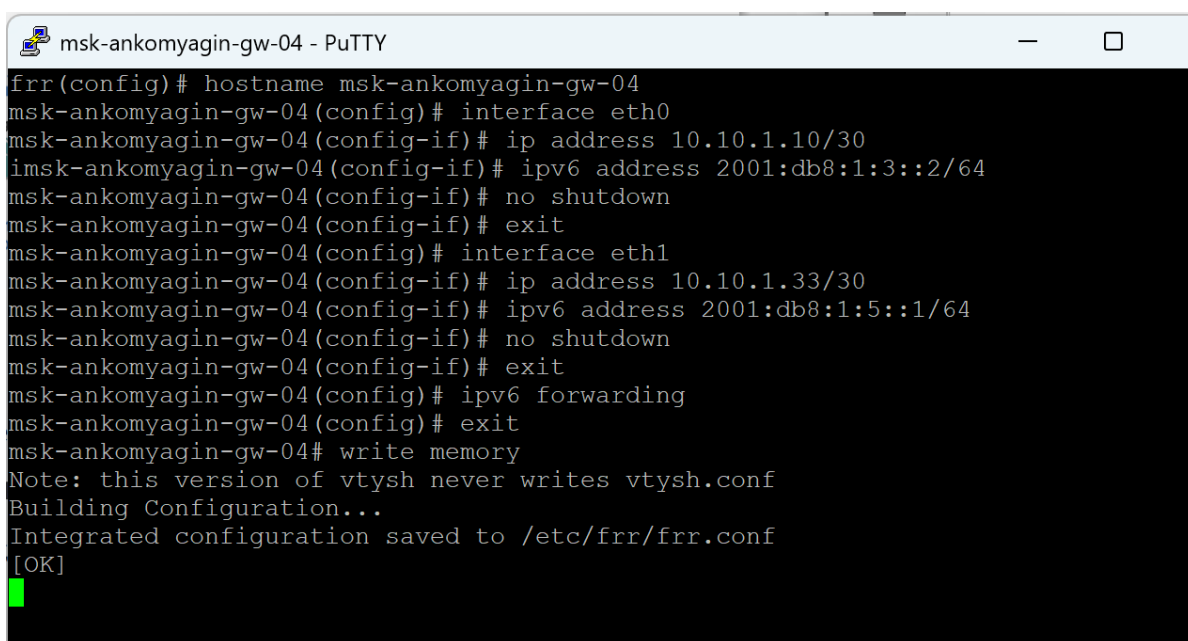
3.5 Динамическая маршрутизация OSPF

(OSPFv2/OSPFv3)

3.5.1 OSPFv2 для IPv4

OSPF — протокол маршрутизации по состоянию канала; в IPv4 используется OSPFv2 и multicast-адреса 224.0.0.5 и 224.0.0.6 для служебного обмена.

На маршрутизаторах настроен OSPFv2 в area 0.0.0.0 с объявлением сетей командой `network` для соответствующих IPv4-подсетей.



```
msk-ankomyagin-gw-04 - PuTTY
frr(config)# hostname msk-ankomyagin-gw-04
msk-ankomyagin-gw-04(config)# interface eth0
msk-ankomyagin-gw-04(config-if)# ip address 10.10.1.10/30
msk-ankomyagin-gw-04(config-if)# ipv6 address 2001:db8:1:3::2/64
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# interface eth1
msk-ankomyagin-gw-04(config-if)# ip address 10.10.1.33/30
msk-ankomyagin-gw-04(config-if)# ipv6 address 2001:db8:1:5::1/64
msk-ankomyagin-gw-04(config-if)# no shutdown
msk-ankomyagin-gw-04(config-if)# exit
msk-ankomyagin-gw-04(config)# ipv6 forwarding
msk-ankomyagin-gw-04(config)# exit
msk-ankomyagin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```

Рис. 3.43: Конфигурация OSPFv2 на msk-ankomyagin-gw-01 (area 0)

```
PC1-ankomyagin - PuTTY
VPCS> ip 10.10.1.98/27 10.10.1.97
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

VPCS> ip 2001:db8:1:1::a/64
PC1 : 2001:db8:1:1::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:db8:1:1::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10096
RHOST:PORT          : 127.0.0.1:10097
MTU                 : 1500

PC2-ankomyagin - PuTTY
VPCS> ip 10.10.1.66/28 10.10.1.65
Checking for duplicate address...
VPCS : 10.10.1.66 255.255.255.240 gateway 10.10.1.65

VPCS> ip 2001:db8:1:6::a/64
PC1 : 2001:db8:1:6::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

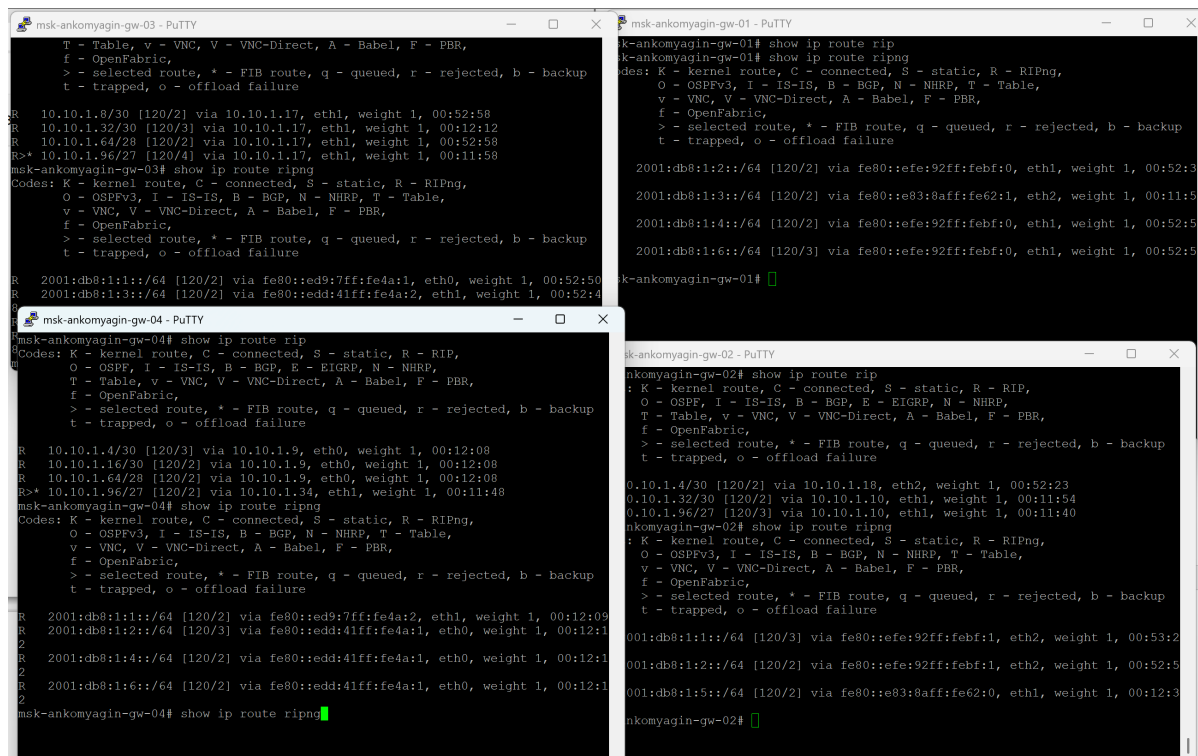
VPCS> show ip

NAME                : VPCS[1]
IP/MASK              : 10.10.1.66/28
GATEWAY              : 10.10.1.65
DNS                  :
MAC                  : 00:50:79:66:68:00
LPORT                : 10098
RHOST:PORT           : 127.0.0.1:10099
MTU                  : 1500

VPCS> █
```

Рис. 3.44: Конфигурация OSPFv2 на msk-ankomyagin-gw-02 (area 0)

Корректность установления соседства и наличие маршрутов проверены командами `show ip ospf neighbor` и `show ip ospf route`.



```
msk-ankomyagin-gw-03 - PuTTY
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
R 10.10.1.8/30 [120/2] via 10.10.1.17, eth1, weight 1, 00:52:58
R 10.10.1.32/30 [120/3] via 10.10.1.17, eth1, weight 1, 00:12:12
R 10.10.1.64/28 [120/2] via 10.10.1.17, eth1, weight 1, 00:52:58
R>* 10.10.1.96/27 [120/4] via 10.10.1.17, eth1, weight 1, 00:11:58
msk-ankomyagin-gw-03# show ip route ripng
Codes: K - kernel route, C - connected, S - static, R - RIPng,
O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
R 2001:db8:1:1::/64 [120/2] via fe80::ed9:7fff:fe4a:1, eth0, weight 1, 00:52:50
R 2001:db8:1:3::/64 [120/2] via fe80::edd:41ff:fe4a:2, eth1, weight 1, 00:52:44

msk-ankomyagin-gw-04 - PuTTY
msk-ankomyagin-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
R 10.10.1.4/30 [120/3] via 10.10.1.9, eth0, weight 1, 00:12:08
R 10.10.1.16/30 [120/2] via 10.10.1.9, eth0, weight 1, 00:12:08
R 10.10.1.64/28 [120/2] via 10.10.1.9, eth0, weight 1, 00:12:08
R>* 10.10.1.96/27 [120/2] via 10.10.1.34, eth1, weight 1, 00:11:48
msk-ankomyagin-gw-04# show ip route ripng
Codes: K - kernel route, C - connected, S - static, R - RIPng,
O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
R 2001:db8:1:1::/64 [120/2] via fe80::ed9:7fff:fe4a:2, eth1, weight 1, 00:12:09
R 2001:db8:1:2::/64 [120/5] via fe80::edd:41ff:fe4a:1, eth0, weight 1, 00:12:12
R 2001:db8:1:4::/64 [120/2] via fe80::edd:41ff:fe4a:1, eth0, weight 1, 00:12:12
R 2001:db8:1:6::/64 [120/2] via fe80::edd:41ff:fe4a:1, eth0, weight 1, 00:12:12
msk-ankomyagin-gw-04# show ip route ripng

msk-ankomyagin-gw-01 - PuTTY
msk-ankomyagin-gw-01# show ip route ripng
Codes: K - kernel route, C - connected, S - static, R - RIPng,
O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
2001:db8:1:2::/64 [120/2] via fe80::efe:92ff:febf:0, eth1, weight 1, 00:52:30
2001:db8:1:3::/64 [120/2] via fe80::e83:8aff:fe62:1, eth2, weight 1, 00:11:55
2001:db8:1:4::/64 [120/2] via fe80::efe:92ff:febf:0, eth1, weight 1, 00:52:55
2001:db8:1:6::/64 [120/3] via fe80::efe:92ff:febf:0, eth1, weight 1, 00:52:59
msk-ankomyagin-gw-01#

msk-ankomyagin-gw-02 - PuTTY
msk-ankomyagin-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
0.10.1.4/30 [120/2] via 10.10.1.18, eth2, weight 1, 00:52:23
0.10.1.32/30 [120/2] via 10.10.1.10, eth1, weight 1, 00:11:54
0.10.1.96/27 [120/3] via 10.10.1.10, eth1, weight 1, 00:11:40
msk-ankomyagin-gw-02# show ip route ripng
Codes: K - kernel route, C - connected, S - static, R - RIPng,
O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure
001:db8:1:1::/64 [120/3] via fe80::efe:92ff:febf:1, eth2, weight 1, 00:53:20
001:db8:1:2::/64 [120/2] via fe80::efe:92ff:febf:1, eth2, weight 1, 00:52:55
001:db8:1:5::/64 [120/2] via fe80::e83:8aff:fe62:0, eth1, weight 1, 00:12:30
msk-ankomyagin-gw-02#
```

Рис. 3.47: Проверка соседей OSPFv2 (`show ip ospf neighbor`)

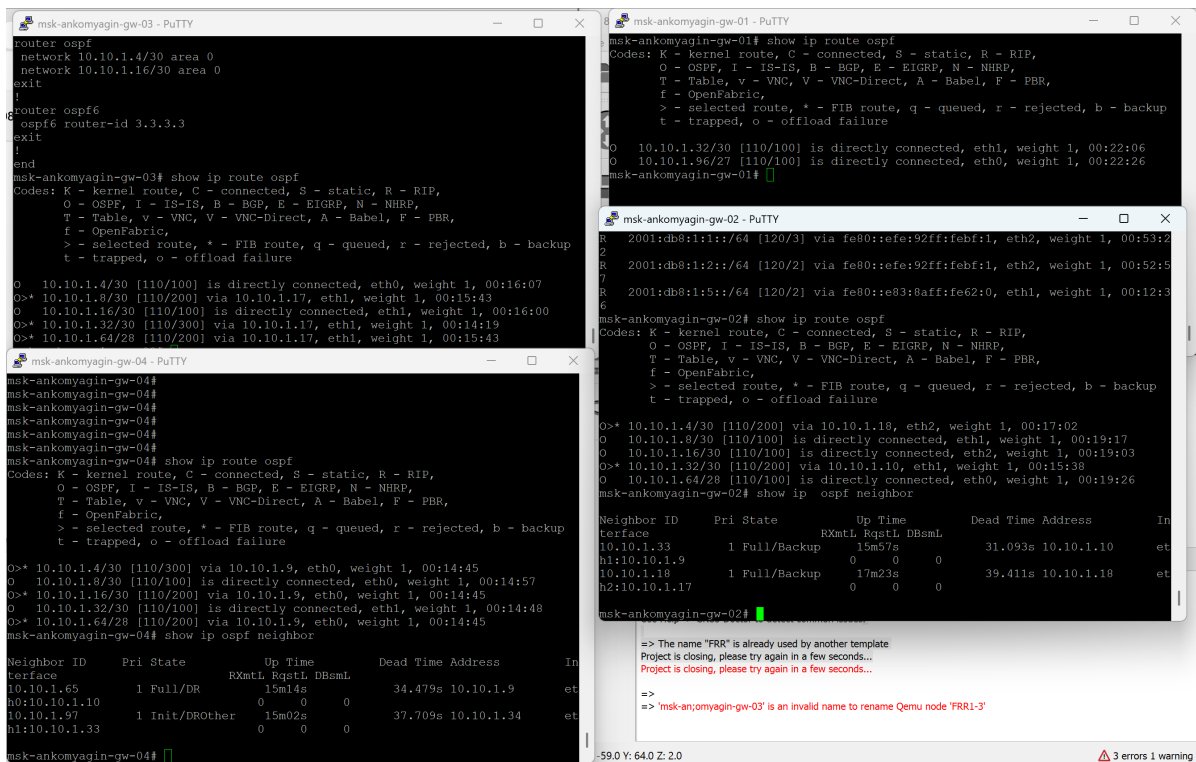


Рис. 3.48: Проверка маршрутов OSPFv2 (show ip ospf route)

Далее выполнены ping/trace PC1 → PC2, затем для проверки сходимости отключился интерфейс на msk-ankomyagin-gw-02 (shutdown eth0) и анализировалось изменение маршрута.

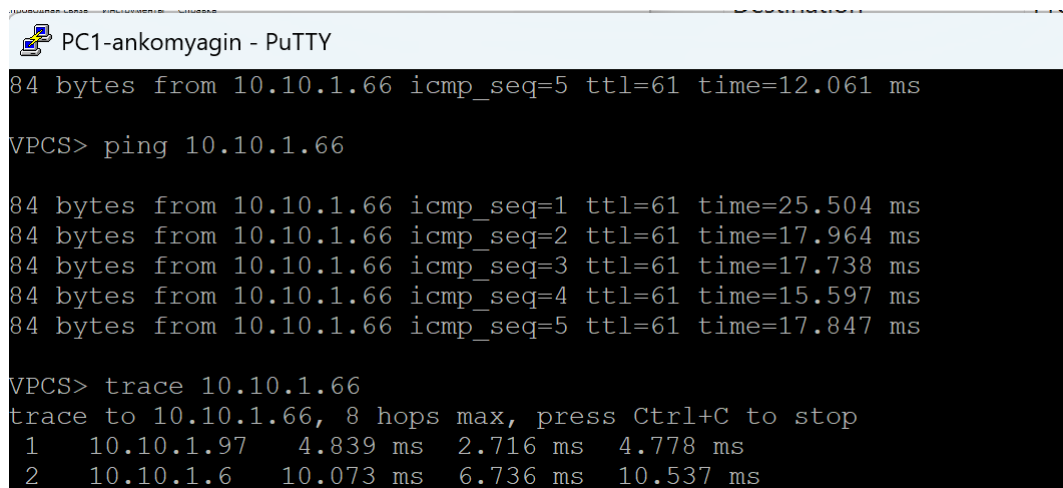


Рис. 3.49: Ping/trace (IPv4) при OSPFv2 до/после отказа линка

В захваченном трафике OSPF можно наблюдать служебные обмены (в т.ч. Hello)

и изменения при потере соседства/перестроении маршрутов.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::ed9:7ff:fe4a:0	ff02::9	RIPng	186	Command Response, Version 1
2	0.301443	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
3	0.498816	10.10.1.97	224.0.0.9	RIPv2	146	Response
4	3.031210	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
5	10.303562	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
6	13.036197	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
7	17.260074	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x268f, seq=1/256, ttl=64 (reply in 8)
8	17.279106	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x268f, seq=1/256, ttl=61 (request in 7)
9	18.285486	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x278f, seq=2/512, ttl=64 (reply in 10)
10	18.302811	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x278f, seq=2/512, ttl=61 (request in 9)
11	19.305210	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x288f, seq=3/768, ttl=64 (reply in 12)
12	19.321494	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x288f, seq=3/768, ttl=61 (request in 11)
13	20.010524	fe80::ed9:7ff:fe4a:0	ff02::9	RIPng	186	Command Response, Version 1
14	20.304078	10.10.1.97	224.0.0.5	OSPF	78	Hello Packet
15	20.323171	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x298f, seq=4/1024, ttl=64 (reply in 16)
16	20.338210	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x298f, seq=4/1024, ttl=61 (request in 15)
17	21.342937	10.10.1.98	10.10.1.66	ICMP	98	Echo (ping) request id=0x2a8f, seq=5/1280, ttl=64 (reply in 18)
18	21.358946	10.10.1.66	10.10.1.98	ICMP	98	Echo (ping) reply id=0x2a8f, seq=5/1280, ttl=61 (request in 17)
19	22.349071	0c:d9:07:4a:00:00	Private_66:68:01	ARP	60	Who has 10.10.1.98? Tell 10.10.1.97
20	22.353404	Private_66:68:01	0c:d9:07:4a:00:00	ARP	60	10.10.1.98 is at 00:50:79:66:68:01
21	23.039866	fe80::ed9:7ff:fe4a:0	ff02::5	OSPF	90	Hello Packet
22	26.503578	10.10.1.97	224.0.0.9	RIPv2	146	Response

Рис. 3.50: Захват трафика OSPF/служебных пакетов на контролируемых линках

4 Выводы

В ходе работы задокументирована топология и адресация стенда, настроен двойной стек IPv4/IPv6 и проверена базовая связность согласно заданию.

Для IPv4 и IPv6 настроены протоколы динамической маршрутизации RIP/RIPng и OSPFv2/OSPFv3, выполнены проверки ping/traceroute и показана реакция сети на отказ интерфейса (перестроение маршрута).

Анализ захваченного трафика позволил выделить служебные рассылки протоколов маршрутизации (например, RIPv2 на 224.0.0.9, RIPng на FF02::9, а также OSPF на 224.0.0.5/224.0.0.6 и FF02::5/FF02::6) и связать их с этапами сходимости маршрутов.